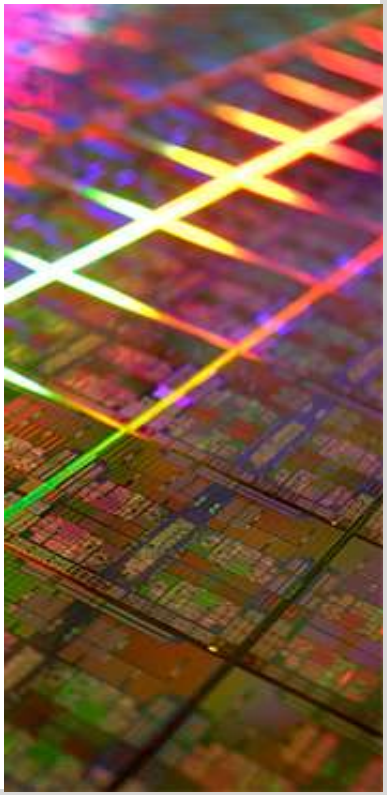
Specialty Chemicals and Gases for Semiconductor Devices Fabrication (Metallization & Planarization Process)

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Ph.D., Tokyo Institute of Technology
Multiple International Awards

23+ Years
Semiconductor Experience

Expertise
CMOS | 3D FinFETs
MEMS | Si Photonics

45+ Publications
Peer-Reviewed Journals

27 Patents
Innovation & IP

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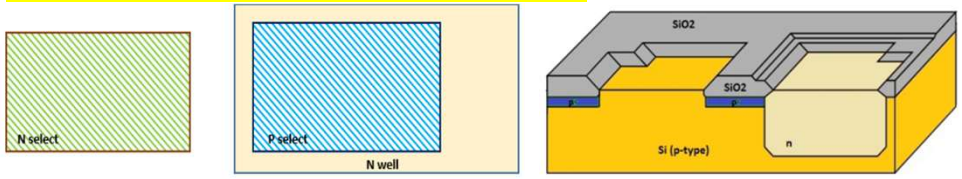
Outline

Section	Topic	Main Focus
1	Introduction to Metallization & CMP	Device integration, interconnects, importance of planarization
2	CMP Fundamentals & Equipment	CMP principles, polishing mechanisms, removal rate
3	CMP Slurry & Specialty Chemicals	Abrasives, oxidizers, complexants, inhibitors, pH, zeta potential
4	Material-Specific CMP Chemistry	Dielectric, STI, tungsten, copper, and other materials
5	CMP Process Control & Defects	Selectivity, endpoint, dishing, erosion, metrology
6	Post-CMP Cleaning	Contamination, cleaning chemistry, cleaning mechanisms
7	Chemical Safety & Waste Management	Chemical hazards, handling, waste segregation and treatment
8	Summary & Key Takeaways	Chemical–mechanical interactions and manufacturing challenges

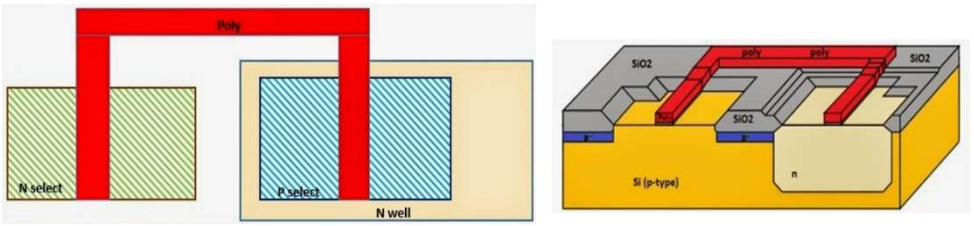
CMOS Inverter process integration

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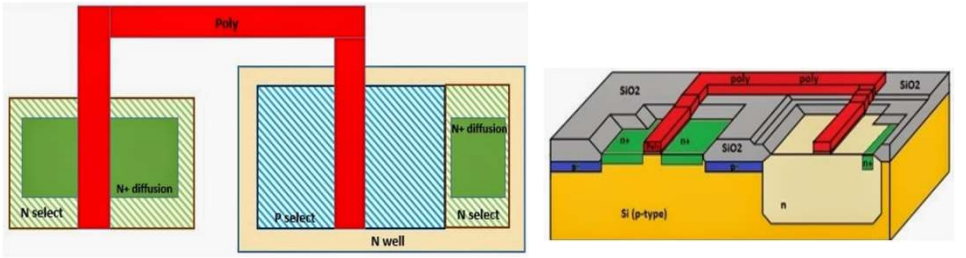
Step 1: N-select, N-well, P-select layers



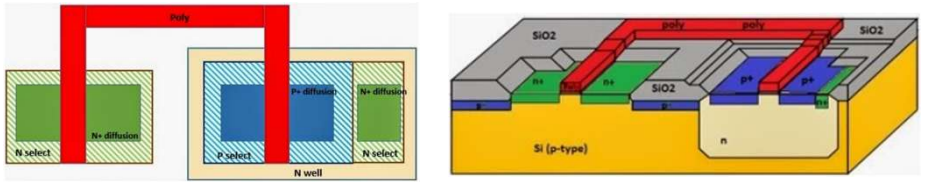
Step 2: Poly-Si layer



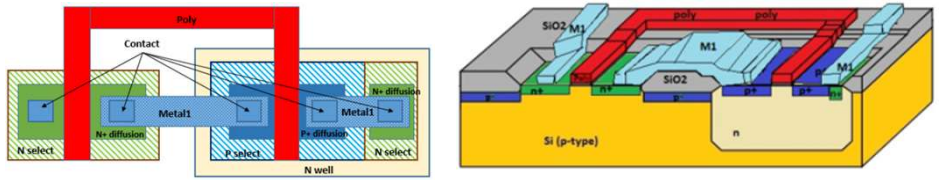
Step 3: N+ Diffusion for NMOS, N-select, N+ diffusion for PMOS body contact

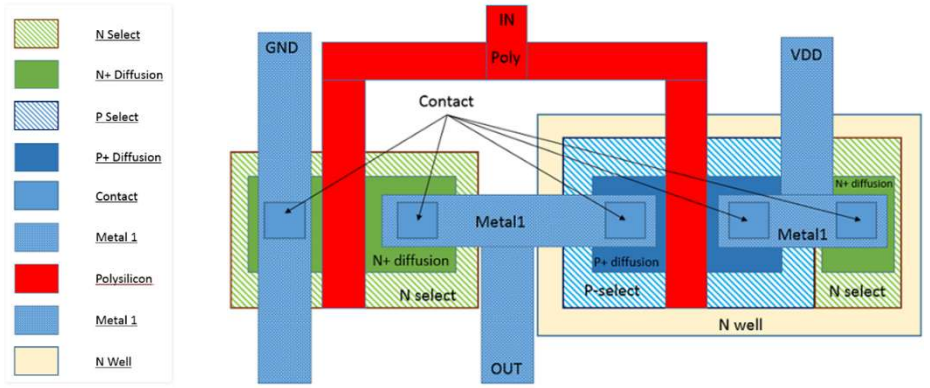


Step 4: P+ Diffusion for PMOS



Step 5: Metal-1 contact layer

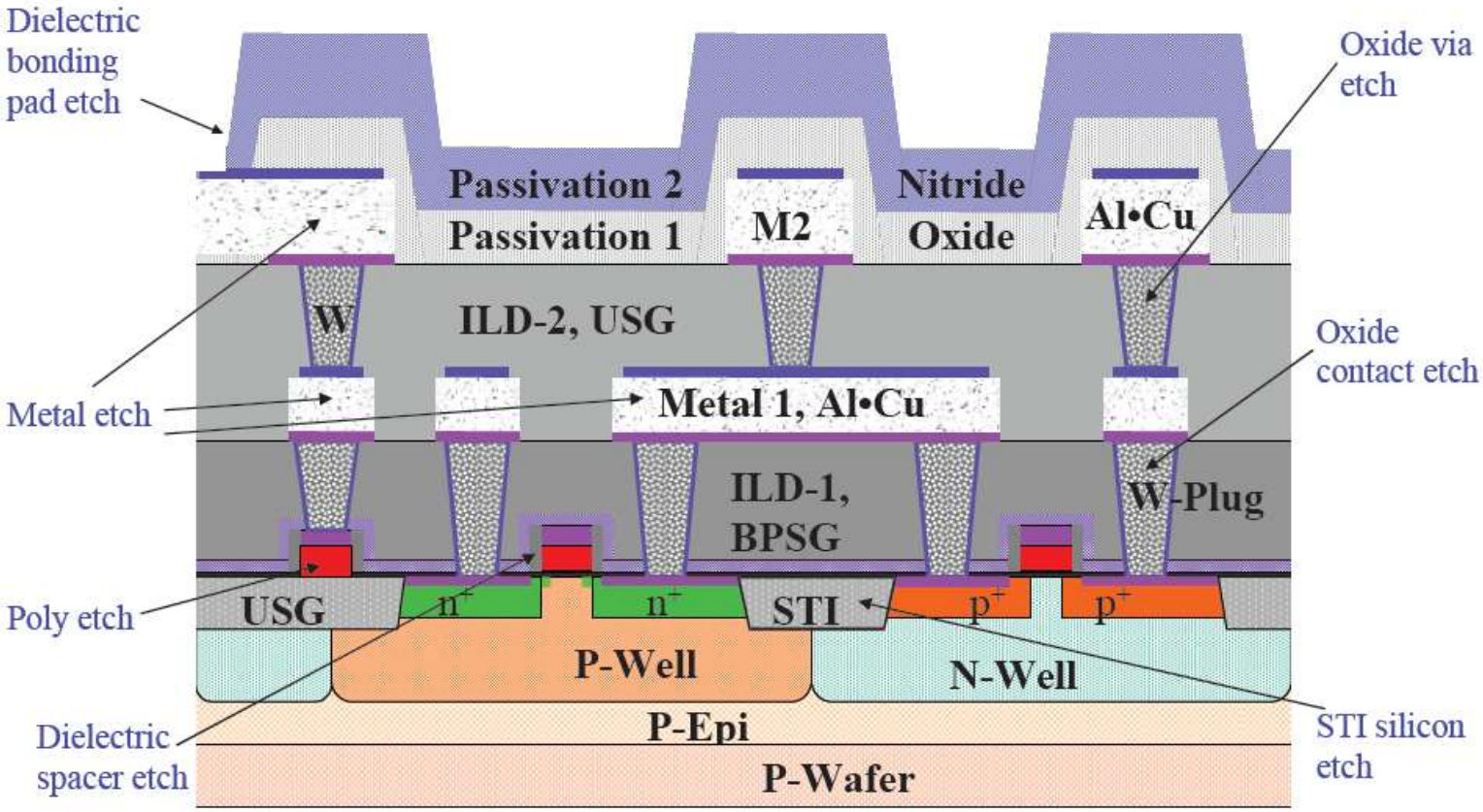




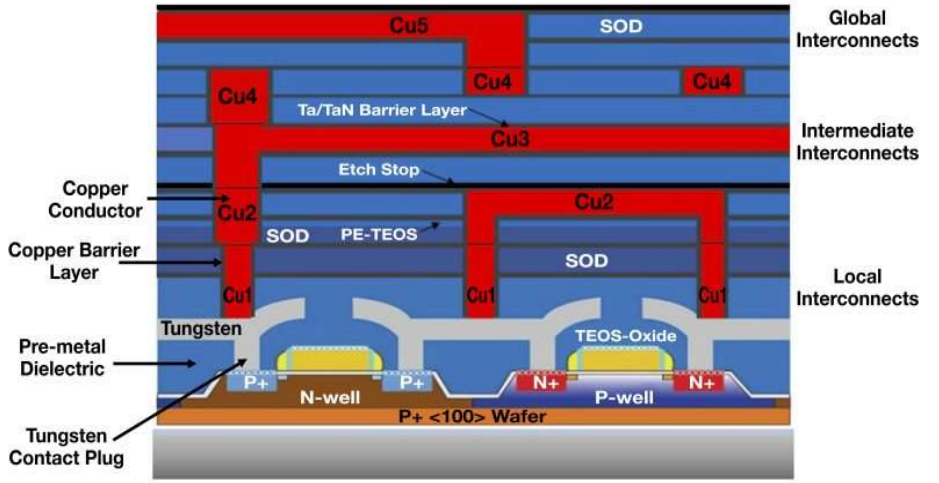
https://www.vlsi-expert.com/2014/#google_vignette

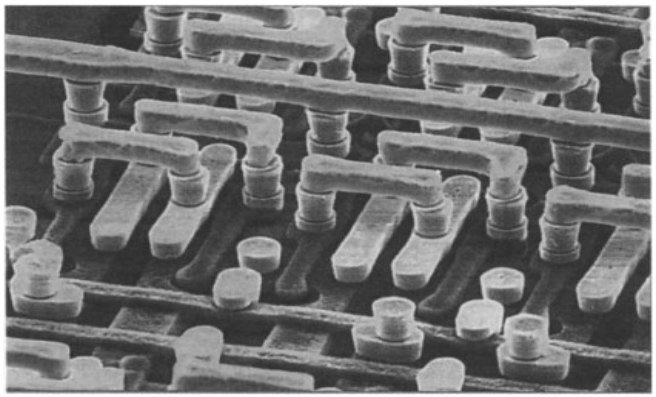
Etching process in MOSFET device

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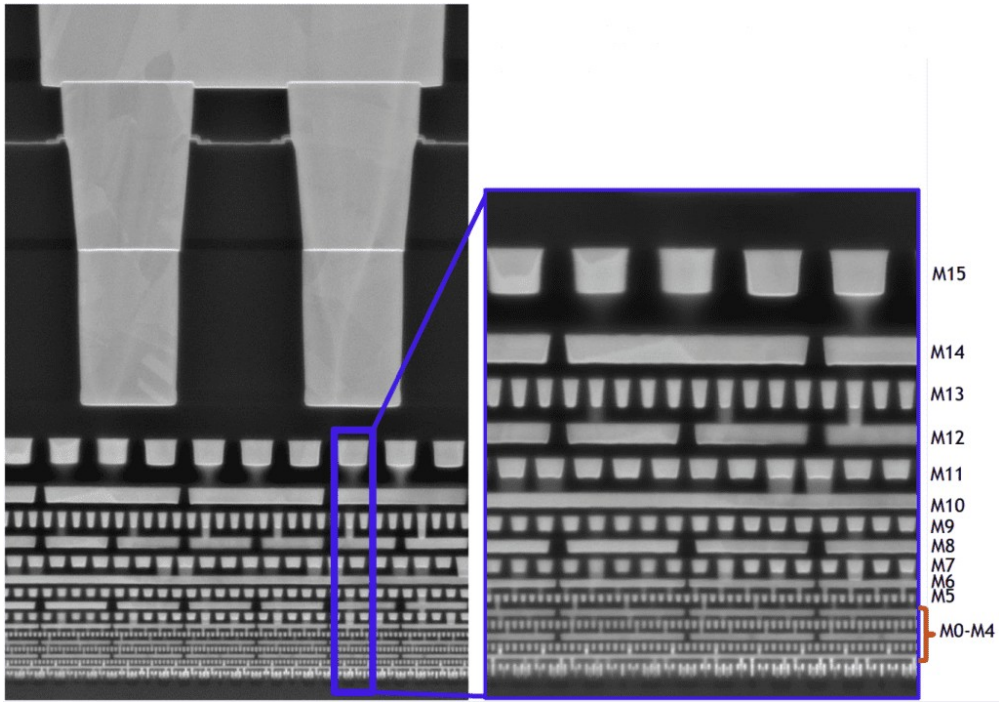


Metal Interconnection (Metal layer + Via)





Layer	Metal	Intel 4 Pitch(nm)
Fin		30
Contacted Gate	W	50
M0	eCu	30
M1	eCu	50
M2	eCu	45
M3	eCu	50
M4	eCu	45
M5	Cu	60
M6	Cu	60
M7	Cu	84
M8	Cu	84
M9	Cu	98
M10	Cu	98
M11	Cu	130
M12	Cu	130
M13	Cu	160
M14	Cu	160
M15	Cu	280
GM0	Cu	1080
GM1	Cu	4000



SEM image of Intel Pentium 4 with 18 metal layers (M1-M18)

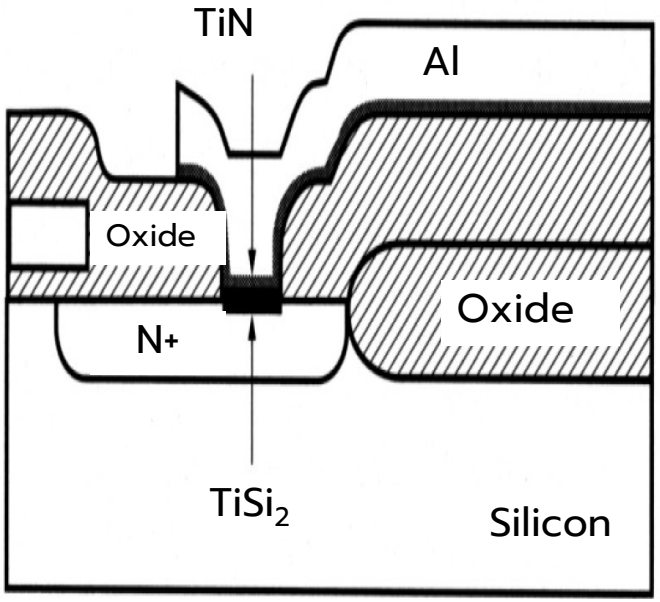
https://www.mks.com/n/metal-thin-films
https://www.semiconductor-digest.com/intel-4-process-drops-cobalt-interconnect-goes-with-tried-and-tested-copper-with-cobalt-liner-cap/

Metal Interconnection

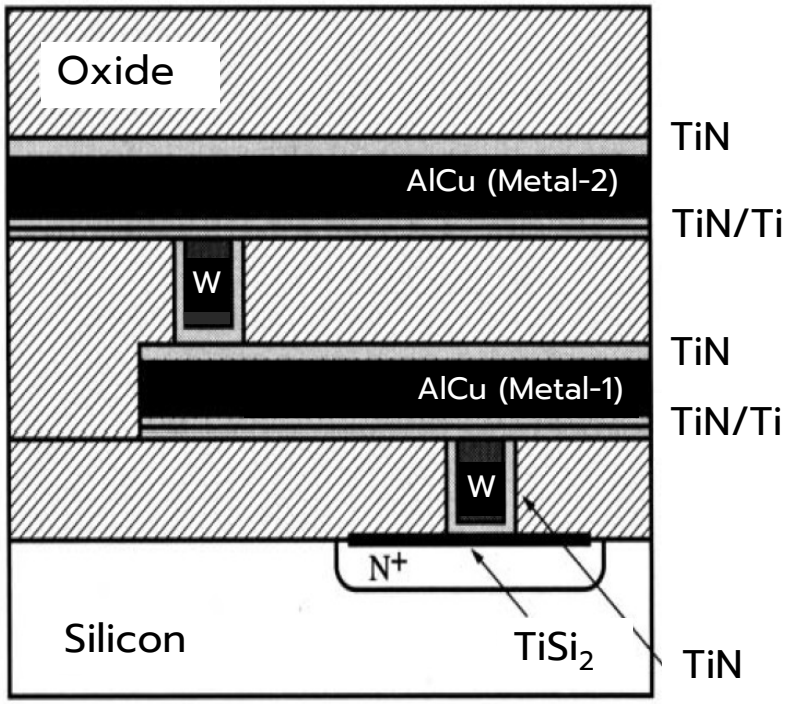
(Al contact, TiN diffusion barrier layer, $TiSi_2$ contact, and W plug)

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Material	Resistivity (Ω-m)	Conductivity (/Ω-m)
Ag	1.59×10^{-8}	6.29×10^{7}
Cu	1.68×10^{-8}	5.95×10^{7}
Au	2.44×10^{-8}	4.10×10^{7}
Al	2.65×10^{-8}	5.81×10^{7}
W	5.6×10^{-8}	1.79×10^{7}
Fe	9.71×10^{-8}	1.03×10^{7}
Pt	10.6×10^{-8}	0.94×10^{7}
Pb	22×10^{-8}	0.45×10^{7}
Mn	48.2×10^{-8}	0.21×10^{7}
(Ni, Cu)	49×10^{-8}	0.20×10^{7}
Hg	98×10^{-8}	010×10^{7}
(Ni, Fe, Cr Alloy)	100×10^{-8}	0.10×10^{7}
Carbon (Graphite)	$3\text{-}60 \times 10^{-5}$	--



Multilayer contact structure with $TiSi_2$/TiN contact/barrier layers between the Si and Al

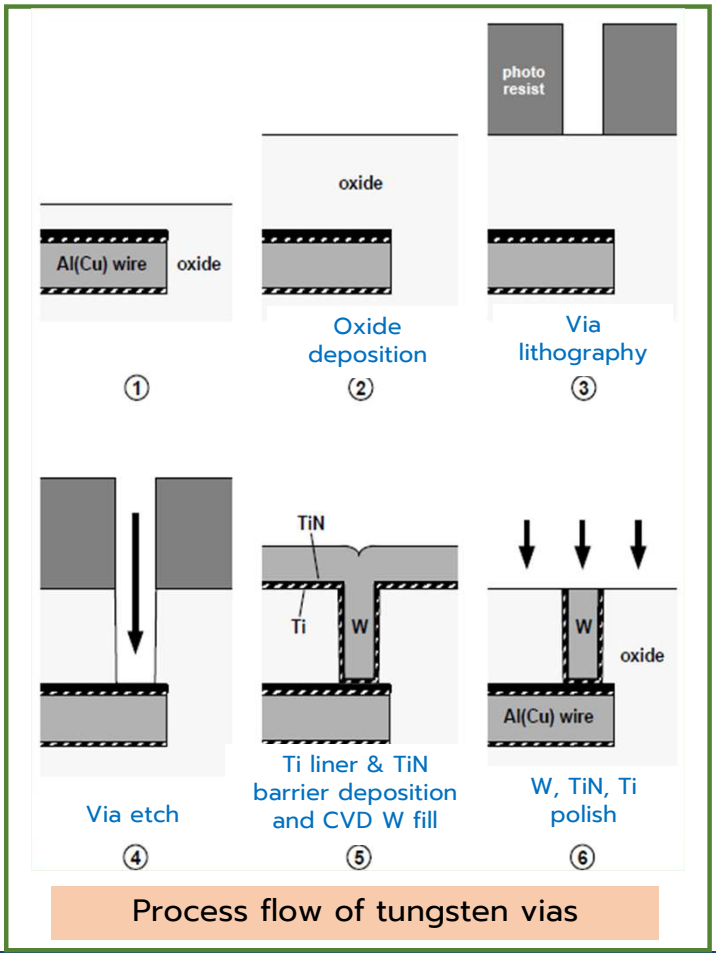


TiN acts like both "Shunt" and Anti-reflective layer

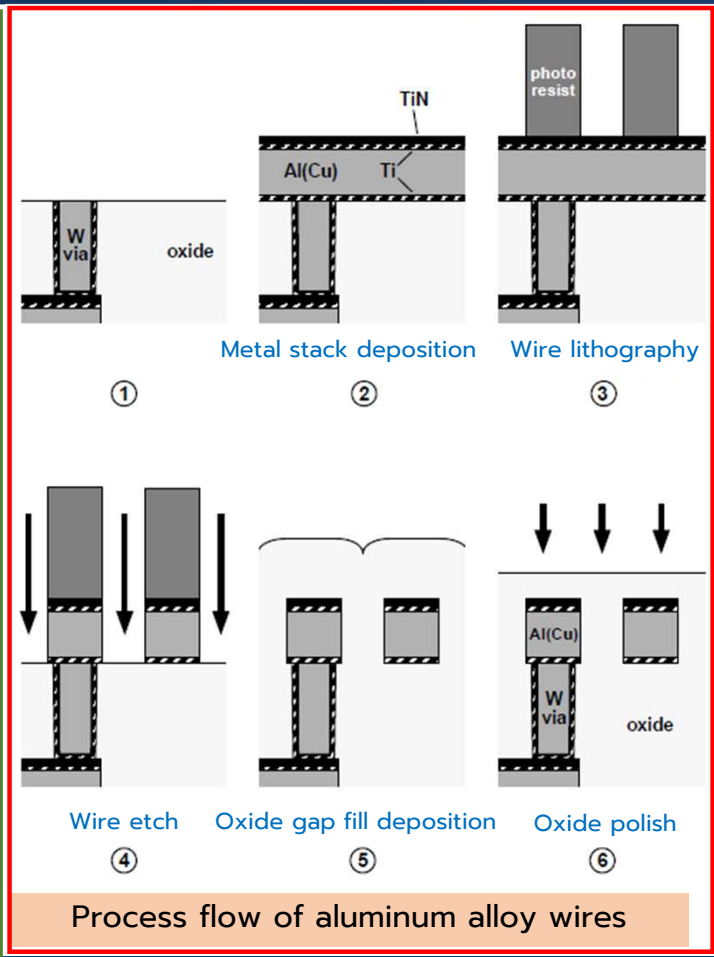
https://www.mks.com/n/metal-thin-films

W vias, Al wires, & Cu Dual Damascene Process

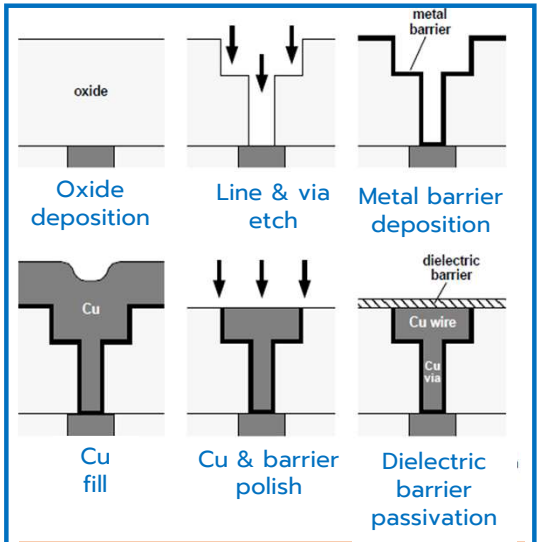
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Process flow of tungsten vias



Process flow of aluminum alloy wires



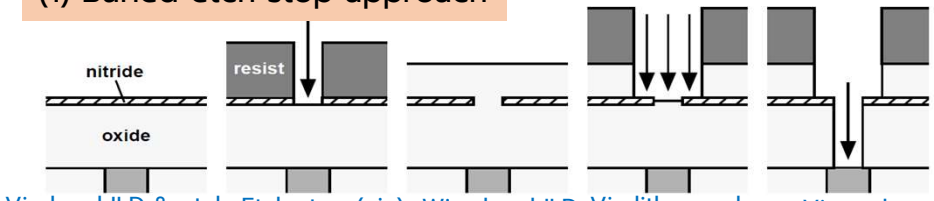
Dual-Damascene process flow for Cu interconnects

DOI: 10.13140/RG.2.1.2425.5766

Cu Dual Damascene methods

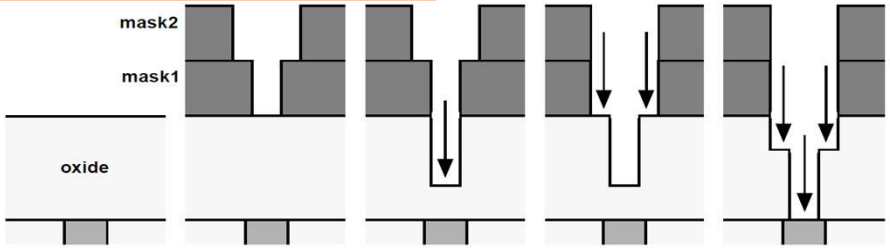
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(1) Buried etch stop approach



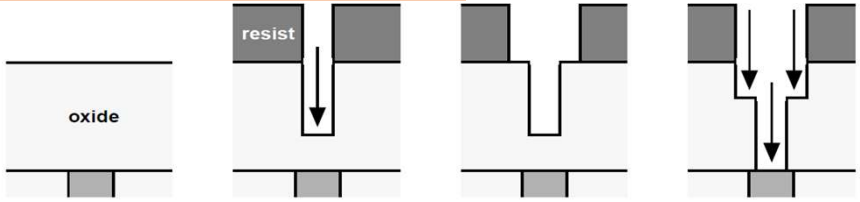
Via-level ILD & etch stop deposition | Etch stop (via) lithography & etch | Wire-level ILD deposition | Via lithography & etch | Via etch

(2) Clustered approach



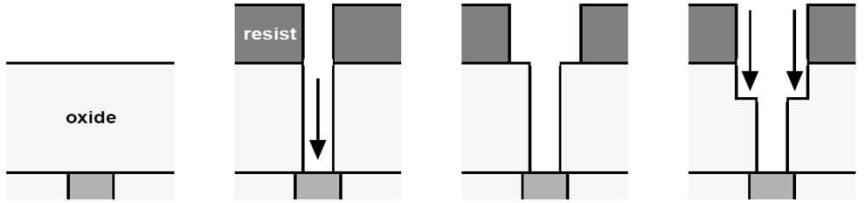
Via-and wire-level ILD deposition | Via-and wire-level lithography | Partial via etch deposition | Via mask etch | Wire and extended via etch

(3) Partial via first approach



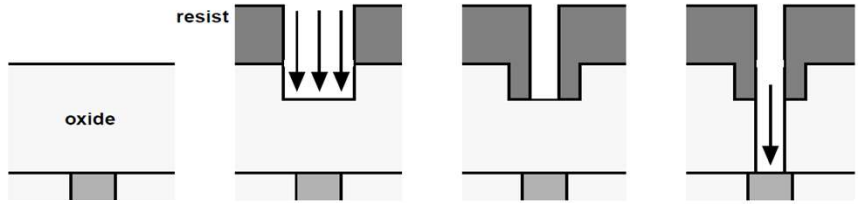
Via-and wire-level ILD deposition | Via lithography & partially etch | Wire lithography | Wire and extended via etch

(4) Full via first approach



Via-and wire-level ILD deposition | Via lithography and etch | Wire lithography | Wire and extended via etch

(5) Line first approach



Via-and wire-level ILD deposition | Wire lithography and etch | Via lithography | Via etch

DOI: 10.13140/RG.2.1.2425.5766

Why Is CMP needed in Semiconductor Manufacturing?

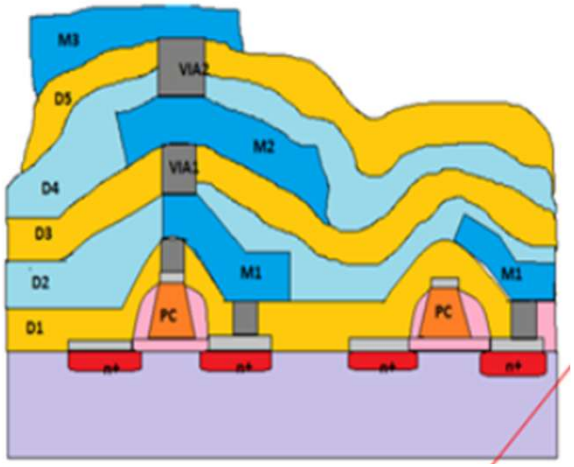
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CMP is essential for achieving a highly planar wafer surface, controlling film thickness, and enabling reliable multilayer device fabrication.

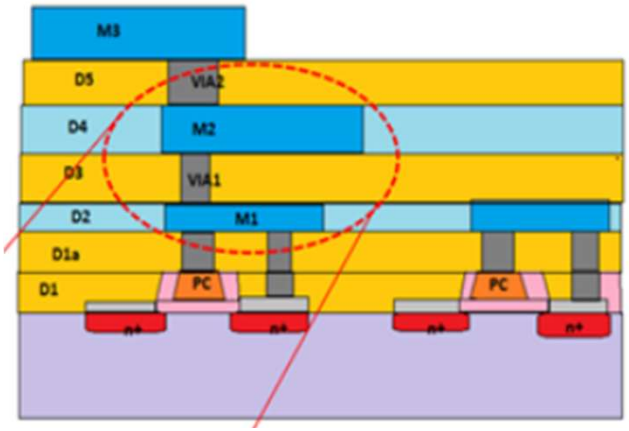
Key Purposes of CMP

1. **Surface Planarization**: Removes surface topography and reduces height variations caused by deposition, etching, and patterning.
2. **Multilayer Integration**: Provides a flat surface for subsequent lithography and thin-film deposition in advanced semiconductor devices.
3. **Improved Lithography**: Minimizes focus variation and depth-of-focus (DOF) limitations, improving pattern resolution and overlay performance.
4. **Selective Material Removal**: Removes excess materials while retaining desired structures, such as Cu interconnects, W plugs, and shallow trench isolation (STI).
5. **Film Thickness Control**: Achieves uniform dielectric and metal film thickness across the wafer.
6. **Improved Device Yield and Reliability**: Reduces topography-related defects, electrical failures, and process variability.

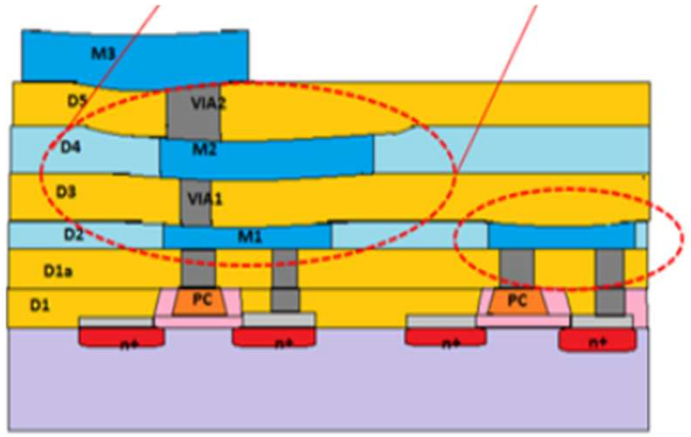
Chemical Mechanical Polishing (CMP)



Chip stack without CMP



Chip stack with CMP
(without CMP defects)



Chip stack with CMP
(with Dishing & Erosion side defects)

https://www.horiba.com/int/semiconductor/process/chemical-mechanical-planarization-cmp/
https://www.mks.com/n/chemical-mechanical-polishing

Chemical Mechanical Planarization (CMP) and Post-CMP Cleaning

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Chemical mechanical polishing (CMP) is a powerful fabrication technique that uses chemical oxidation and mechanical abrasion to remove material and achieve very high levels of planarity.

CMP:

- Combines **chemical reactions (oxidation)** and **mechanical abrasion** to achieve selective material removal and high surface planarity.
- Uses specially formulated **slurries containing chemicals and abrasive particles**.
- Widely applied in semiconductor manufacturing for **surface planarization** and device structure formation.

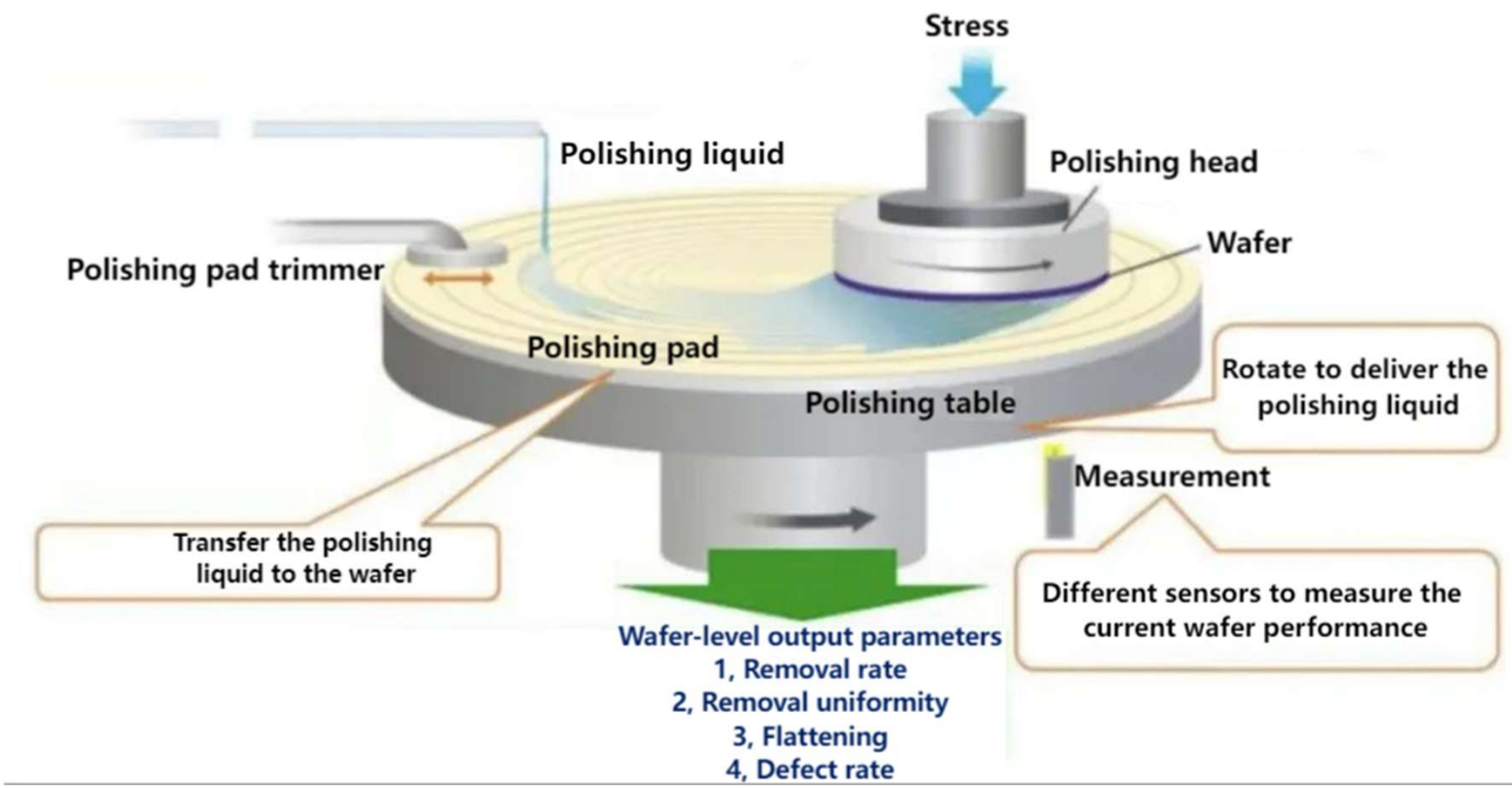
Post-CMP Cleaning:

- Removes **abrasive particles, polishing byproducts, organic residues, and metallic contaminants** remaining after CMP.
- Prevents defects caused by slurry particles and contaminants adhering to wafer surfaces.
- Ensures wafer cleanliness while minimizing **scratches, watermarks, surface roughness, corrosion, and dielectric property changes**.

https://www.horiba.com/ind/semiconductor/applications/chemical-mechanical-planarization-cmp/

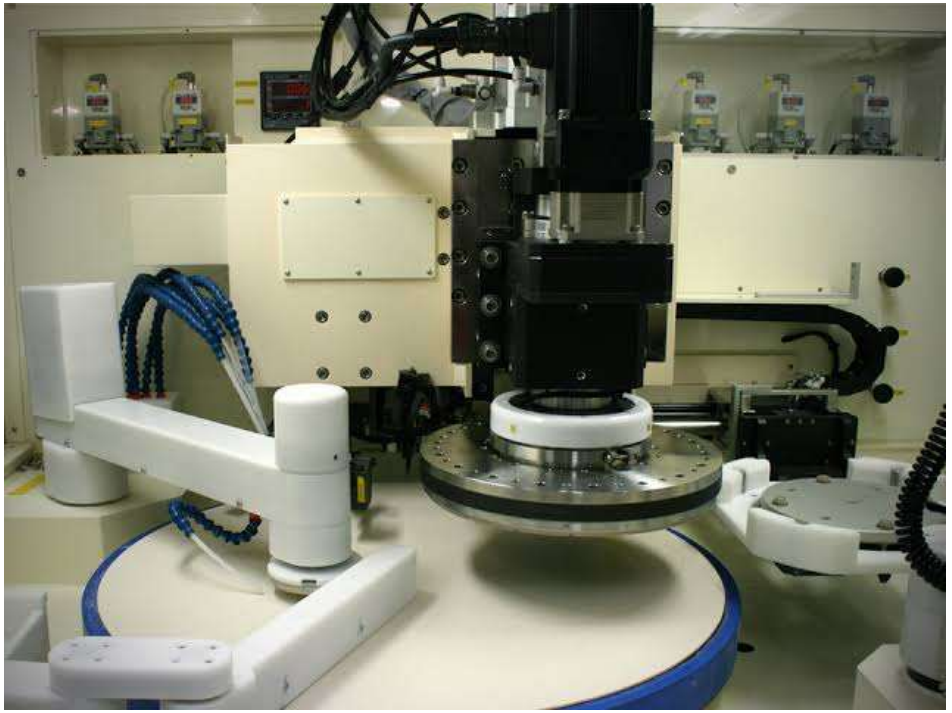
CMP Equipment and Operating Principles

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CMP Machine





https://www.ebara.com.sg/precision-machinery-products/semiconductor-manufacturing-system/chemical-mechanical-polishing-cmp-f-rex300x

CMP Removal Rate & Preston's Equation

Chemical Mechanical Polishing (CMP) combines chemical reactions and mechanical abrasion to remove materials and achieve wafer surface planarization.

$$RR = K_p \times P \times V$$

Parameter	Description	Typical Unit
RR	Material removal rate	nm/min or μm/min
K_p	Preston coefficient (process-dependent)	nm/(min·kPa·m/s)
P	Applied polishing pressure	kPa or psi
V	Relative velocity between wafer and polishing pad	m/s

Key Factors Affecting CMP Removal Rate

Mechanical Factors

- Downforce / pressure
- Pad and platen velocity
- Pad hardness and conditioning
- Abrasive particle size and concentration

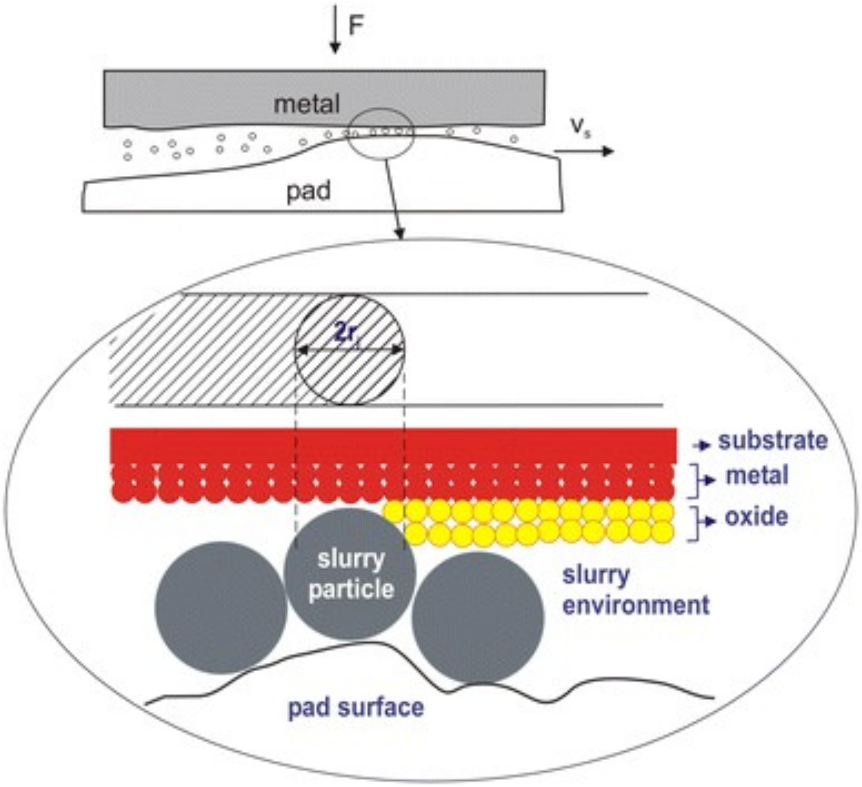
Chemical Factors

- Slurry pH
- Oxidizers (e.g., H_2O_2)
- Complexing agents and inhibitors
- Material surface chemistry

CMP Removal Rate & Preston's Equation

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- Increasing polishing pressure or relative velocity generally increases removal rate.
- The Preston coefficient depends on slurry chemistry, pad properties, wafer material, and process conditions.
- Preston's equation is an empirical approximation; real CMP may exhibit nonlinear behavior.
- CMP optimization requires balancing removal rate, selectivity, uniformity, dishing, erosion, and defectivity.



Major CMP slurry components

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CMP slurry is a liquid chemical mixture containing fine abrasive particles and chemical additives, used during Chemical Mechanical Planarization (CMP) to remove material from a wafer surface and achieve a highly flat, smooth surface. CMP slurry combines two mechanisms:

1. **Chemical action**: Chemicals react with the wafer surface, forming a modified layer that is easier to remove.
2. **Mechanical action**: Abrasive particles, together with the polishing pad, mechanically remove the reacted material.

Component	Function	Typical chemicals/materials
Abrasive	Mechanical removal	SiO_2, CeO_2, Al_2O_3
Oxidizer	Oxidizes/softens metal surface	H_2O_2, $Fe(NO_3)_3$, KIO_3
Complexing/Chelating agent	Keeps dissolved metal ions soluble	Glycine, citric acid
Corrosion inhibitor	Prevents excessive metal dissolution	Benzotriazole (BTA)
pH adjuster/buffer	Controls surface reaction and particle charge	KOH, NH_4OH, HNO_3, organic acids
Surfactant	Wetting / particle interaction / defect control	Proprietary surfactants
Dispersant	Prevents abrasive agglomeration	Polymers / polyelectrolytes
Biocide	Controls microbial growth	Proprietary
DI/UPW	Solvent/carrier	Ultrapure water

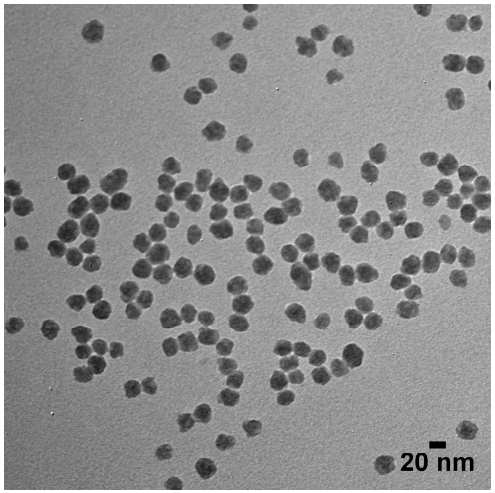
Chemicals Used in CMP Process

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CMP chemicals are mainly divided into abrasives, oxidizers, complexing agents, corrosion inhibitors, pH adjusters, surfactants, and post-CMP cleaning chemicals.

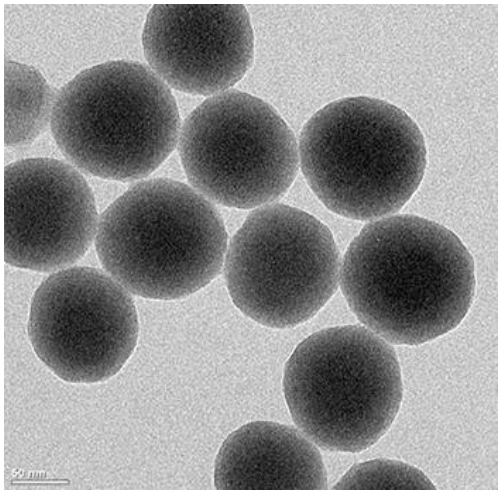
Category	Chemical	Formula	Main function
Abrasive	Colloidal silica	SiO_2	Mechanical removal and polishing
	Cerium oxide (ceria)	CeO_2	Oxide polishing, high SiO_2 selectivity
	Aluminum oxide (alumina)	Al_2O_3	Metal polishing
Oxidizer	Hydrogen peroxide	H_2O_2	Oxidizes Cu, W and other metals
	Ferric nitrate	$Fe(NO_3)_3$	Oxidation, particularly in W CMP
	Potassium iodate	KIO_3	Oxidizer for certain metal CMP processes
Complexing agent	Glycine	$C_2H_5NO_2$	Complexes dissolved metal ions
	Citric acid	$C_6H_8O_7$	Metal-ion complexation
	EDTA	$C_{10}H_{16}N_2O_8$	Metal chelation
Corrosion inhibitor	Benzotriazole (BTA)	$C_6H_5N_3$	Prevents Cu corrosion
	Tolyltriazole (TTA)	$C_7H_7N_3$	Protects Cu surfaces
pH adjuster	Potassium hydroxide	KOH	Adjusts alkaline slurry pH
	Ammonium hydroxide	NH_4OH (aq.)	Adjusts alkaline slurry pH
	Nitric acid	HNO_3	Adjusts acidic slurry pH
	Sulfuric acid	H_2SO_4	Adjusts acidic slurry pH
Surfactant	Polyethylene glycol (PEG)	$H(OCH_2CH_2)_nOH$	Dispersion and surface wetting
Post-CMP cleaner	Tetramethylammonium hydroxide	TMAH	Removes particles and residues in formulated cleaners
	Hydrofluoric acid	HF	Removes oxide residues in selected processes
	Citric acid	$C_6H_8O_7$	Removes metal contamination
	Ammonium hydroxide	NH_4OH (aq.)	Alkaline cleaning and particle removal

Types of CMP Slurry



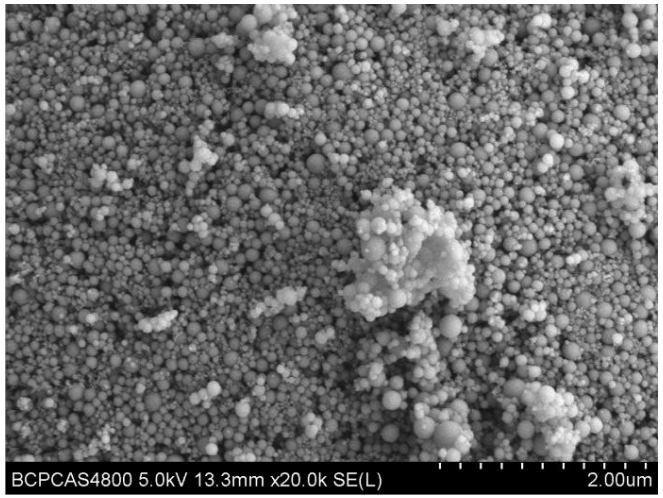
Silica-based slurry (SiO_2)

Commonly used for Si, SiO_2, Cu, W, and other materials. Offers controllable removal rates and relatively low scratching when properly formulated.



Ceria-based slurry (CeO_2)

Widely used for shallow trench isolation (STI) oxide CMP. Surface chemical interactions can provide high oxide removal with controlled nitride selectivity.



Alumina-based slurry (Al_2O_3)

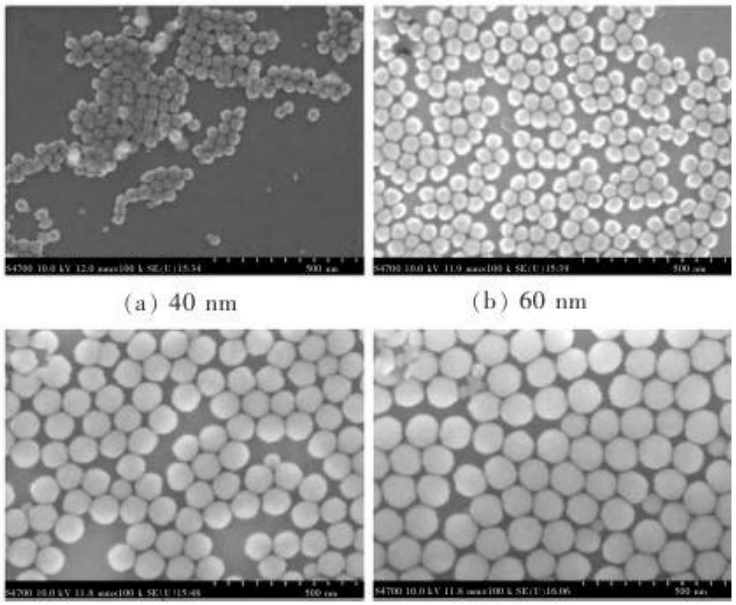
Used in selected metal CMP processes, including tungsten polishing. Its hardness can improve abrasion but requires careful scratch control.

CMP Slurry: Particle Size and Zeta Potential

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Slurry particle size and zeta potential are critical parameters controlling CMP removal rate, surface quality, particle dispersion stability, and defectivity.

Parameter	Effect on CMP Performance
Particle Size	Typically ~20–200 nm, depending on abrasive material and application
Small Particles	Generally reduce scratching; may lower mechanical removal efficiency
Large Particles	Can increase mechanical removal but raise scratch risk
Particle Size Distribution (PSD)	Narrow distribution improves process consistency
Agglomeration	Large particle clusters can cause scratches and defects



(a) 40 nm (b) 60 nm

(c) 90 nm (d) 120 nm

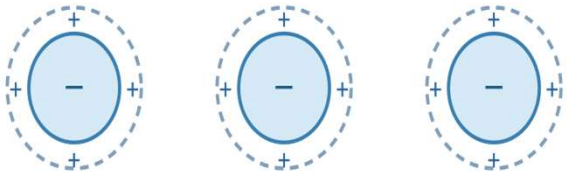
CMP Slurry: Particle Size and Zeta Potential

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Zeta potential is the electrokinetic potential at the slipping plane of particles suspended in a liquid. It influences electrostatic repulsion, colloidal stability, and particle–wafer interactions.

Impact on CMP Performance

- Removal Rate (RR): Particle size, concentration, and surface charge affect mechanical and chemical interactions.
- Defectivity: Agglomerated particles and large-particle tails increase scratching risk.
- Slurry Stability: Sufficient electrostatic or steric repulsion helps prevent particle aggregation.
- Post-CMP Cleaning: Particle–wafer electrostatic interactions influence particle adhesion and removal.



Charged particles and surrounding ion layers
Electrostatic repulsion helps prevent agglomeration

Zeta Potential	General Dispersion Behavior
ζ ≤ -30 mV or ≥ +30 mV	Often good electrostatic stability
-30 to -10 mV or +10 to +30 mV	Moderate or formulation-dependent stability
-10 to +10 mV	Higher aggregation risk without other stabilization

Typical CMP Slurry Compositions

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CMP type	Abrasive concentration	Oxidizer concentration	Typical pH
Si CMP	Silica: 1–10 wt%	Usually none added	9–11
SiO_2 CMP	Silica: 2–15 wt%	Usually none added	9–11
STI oxide CMP	Ceria: 0.1–5 wt%	Usually none added	4–10
Cu CMP	Silica: 0.1–5 wt%	H_2O_2: 0.1–5 wt%	3–8
W CMP	Silica: 1–5 wt%	H_2O_2: 0.1–5 wt%	2–5
SiC CMP	Silica: 1–10 wt%	Formulation-dependent	2–11

CMP Chemicals by Semiconductor Material

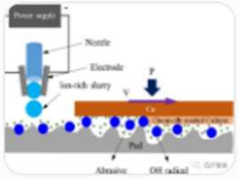
CMP application	Typical abrasive	Key chemicals	Main objective
Silicon (Si)	Colloidal silica	KOH, NH_4OH, organic additives	Silicon planarization
Silicon dioxide (SiO_2)	Silica, ceria	KOH, NH_4OH, dispersants	Dielectric planarization
Silicon nitride (Si_3N_4)	Silica, ceria	pH modifiers, selectivity additives	Nitride removal or controlled stopping
Copper (Cu)	Silica, alumina	H_2O_2, glycine, BTA	Cu removal and planarization
Tungsten (W)	Silica, alumina	H_2O_2, $Fe(NO_3)_3$, organic acids	W plug planarization
Tantalum/TaN	Silica, alumina	Oxidizers, complexing agents, pH modifiers	Barrier removal
Titanium/TiN	Silica, alumina	H_2O_2, organic acids	Metal/barrier planarization
Cobalt (Co)	Silica	H_2O_2, complexants, corrosion inhibitors	Co interconnect planarization
Ruthenium (Ru)	Silica, alumina	Formulated oxidizers, complexants	Ru interconnect planarization
Silicon carbide (SiC)	Silica, ceria, alumina	H_2O_2, $KMnO_4$ or other oxidizers	Surface finishing and defect reduction

Oxidation and Complexation in Metal CMP

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Metal CMP combines chemical oxidation, complexation, and mechanical abrasion to achieve controlled metal removal and surface planarization.

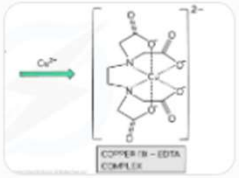
Oxidation modifies the metal surface, complexation helps remove dissolved metal species, and mechanical abrasion continuously renews the surface.



Step 1 — Oxidation

Oxidizing agents such as H_2O_2 convert the metal surface into oxides or oxidized species, making it more susceptible to removal.

$$M \rightarrow M^{n+} + ne^-$$

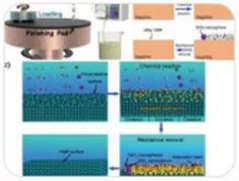


Step 2 — Complexation

Complexing agents bind dissolved metal ions, promoting their solubility and controlling dissolution.

$$M^{n+} + L \rightleftharpoons ML^{n+}$$

Shown for a neutral ligand L; complex charge depends on ligand chemistry.



Step 3 — Mechanical Removal

Abrasive particles and polishing pad contact remove the chemically modified surface layer, exposing fresh metal for continued oxidation.

Typical Chemicals Used in Metal CMP

Function	Typical Chemicals	Purpose
Oxidizer	H_2O_2, $Fe(NO_3)_3$, KIO_3	Promote metal oxidation
Complexing Agent	Glycine, citric acid, EDTA	Bind metal ions and control dissolution
Corrosion Inhibitor	Benzotriazole (BTA), triazoles	Suppress uncontrolled corrosion, particularly for Cu
Abrasive	Silica (SiO_2), alumina (Al_2O_3)	Mechanically remove modified surface layers
pH Adjuster	KOH, NH_4OH, HNO_3	Control surface chemistry and reaction kinetics

Key Process Challenges

- Excessive oxidation: May increase corrosion, pitting, and unwanted metal dissolution.
- Excessive complexation: May increase static etch rate and cause dishing.
- Insufficient oxidation: Can reduce removal rate and polishing efficiency.
- Chemical balance: Oxidizers, complexants, inhibitors, and abrasives must be optimized together.

Surface Chemistry in Dielectric CMP

Dielectric CMP relies on surface hydration, chemical bond formation, and mechanical abrasion to remove SiO_2 and other dielectric materials while achieving high planarization efficiency and selectivity.

Chemical Mechanisms of Dielectric CMP

Step 1 — Surface Hydrolysis and Hydration

Water reacts with Si–O–Si bonds on the oxide surface, forming silanol (Si–OH) groups.

$$Si - O - Si + H_2O \rightleftharpoons 2Si - OH$$

Step 2 — Abrasive–Surface Interaction

Silica (SiO_2) or ceria (CeO_2) particles interact with the hydrated dielectric surface.

- Silica: Silanol interactions and mechanical abrasion
- Ceria: Strong chemical affinity toward SiO_2 through interfacial Ce–O–Si bonding

Step 3 — Mechanical Removal

Polishing pad pressure and relative motion facilitate removal of the chemically modified surface layer.

Typical Chemicals Used in Dielectric CMP

Chemical / Material	Function
Colloidal Silica (SiO_2)	Abrasive for oxide planarization
Ceria (CeO_2)	Reactive abrasive for oxide CMP, particularly STI
KOH / NH_4OH	Adjust pH and promote silica surface dissolution
Organic Acids / Buffers	Control slurry pH and surface chemistry
Surfactants / Dispersants	Stabilize abrasive particles and reduce defects
Selectivity Additives	Control oxide-to-Si_3N_4 removal selectivity

Surface Chemistry and CMP Selectivity

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Key Process Challenges

- Oxide removal rate: Optimize hydration, slurry pH, and abrasive chemistry.
- Selectivity: Control SiO_2 removal relative to Si_3N_4 and other materials.
- Defectivity: Minimize scratches, particle residues, and surface damage.
- Planarity: Reduce oxide dishing and pattern-density-dependent erosion.

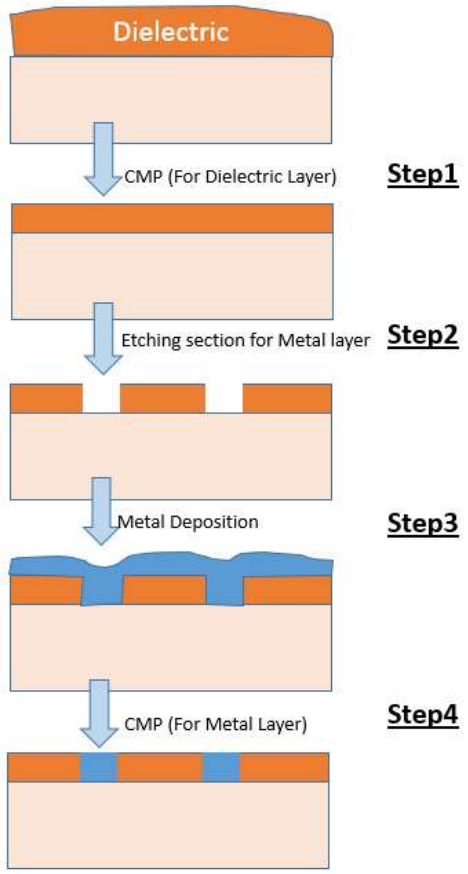
Parameter	Effect on CMP Performance
Slurry pH	Controls surface charge, silica dissolution, and abrasive interactions
Surface Hydration	Facilitates chemical modification and removal
Zeta Potential	Influences particle–wafer attraction, repulsion, and dispersion stability
Abrasive Chemistry	Ceria provides stronger chemical interactions with oxide than silica
Additives	Control selectivity, removal rate, and defectivity

Dielectric Planarization CMP
Oxide Surface Planarization

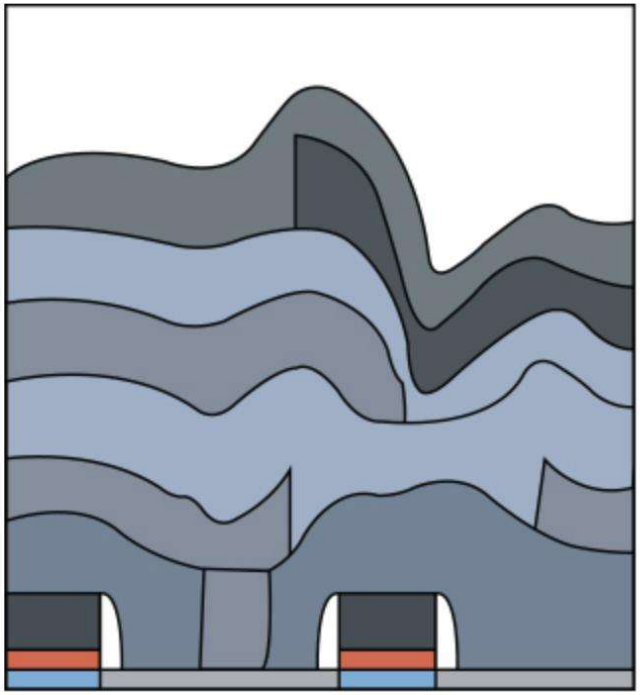
Remove excess dielectric material and eliminate surface topography to achieve a highly planar wafer surface for subsequent lithography, thin-film deposition, and multilayer semiconductor fabrication.

Process Principle

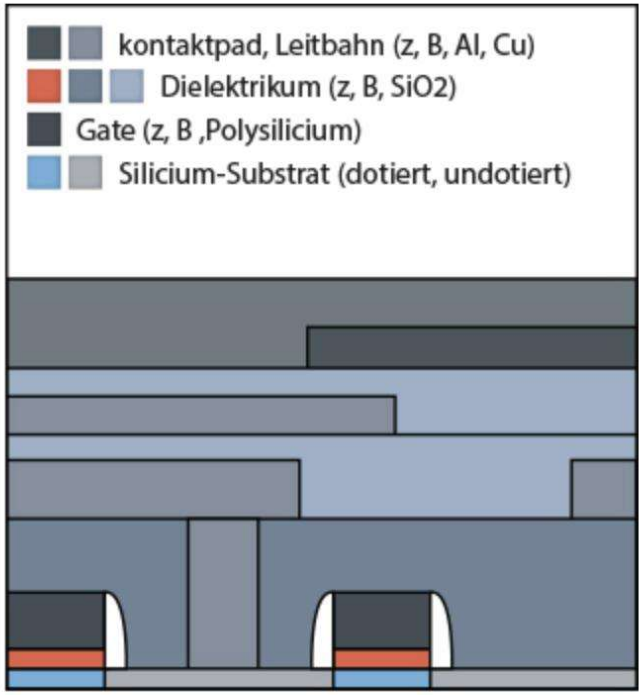
- **Dielectric Deposition**: Deposit SiO_2-based dielectric films (e.g., PECVD SiO_2, TEOS oxide, or HDP-CVD oxide) over patterned structures, creating surface topography.
- **CMP Planarization**: Remove raised dielectric regions through chemical reactions and mechanical abrasion.
- **Surface Leveling**: Higher regions experience greater initial material removal, reducing height variations across the wafer.
- **Thickness Control**: Control the final dielectric thickness, within-wafer uniformity, and surface planarity.



Dielectric Planarization CMP
Oxide Surface Planarization



(CMP 공정 X)



(CMP 공정 O)

Typical Dielectric CMP Materials and Chemicals

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Key Process Challenges

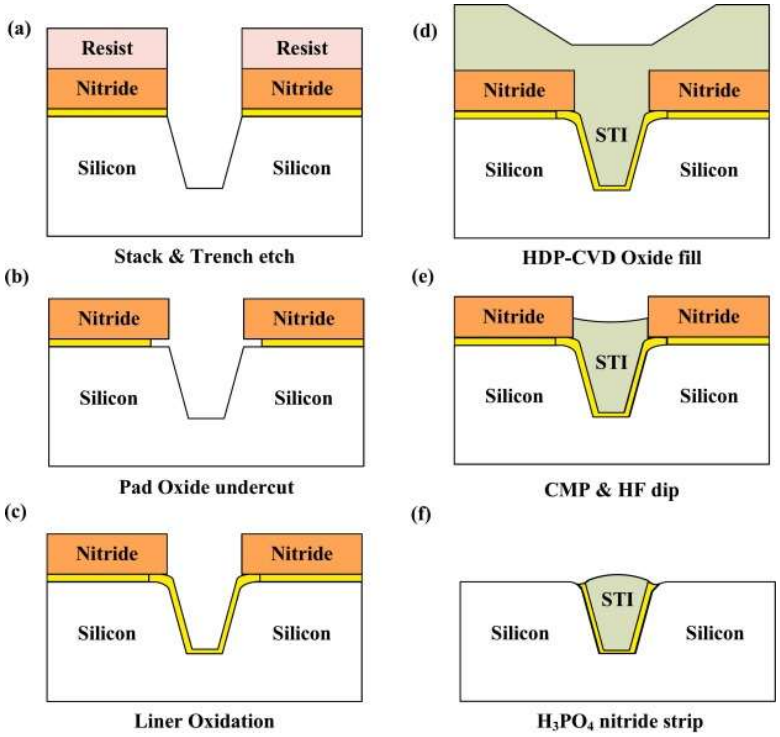
- Within-wafer non-uniformity: Variation in dielectric removal rate and final film thickness.
- Pattern-dependent planarization: Different polishing rates between dense and isolated structures.
- Overpolishing: Excessive dielectric removal, reducing insulation thickness.
- Surface defects: Scratches, residual abrasive particles, and surface roughness.
- Film damage: Moisture uptake or chemical damage in sensitive low-k dielectric materials.

Component	Material / Chemical	Function
Film to polish	SiO_2, TEOS oxide, doped SiO_2	Dielectric planarization
Abrasive	SiO_2 (silica), CeO_2 (ceria)	Mechanical material removal
Slurry chemistry	Alkaline or formulated aqueous solutions	Promote surface reactions and oxide removal
pH adjuster	KOH, NH_4OH, organic bases	Control pH and slurry stability
Additives	Dispersants, surfactants	Improve particle stability and defect control
Cleaning chemicals	DI water, dilute alkaline or formulated cleaners	Remove residual slurry and particles

STI CMP – Removal of Overburden Oxide

Remove excess SiO_2 deposited during Shallow Trench Isolation (STI) fabrication while maintaining oxide-filled trenches and a planar wafer surface.

- Oxide Deposition: Fill etched silicon trenches with SiO_2 (e.g., HDP-CVD or flowable oxide), leaving excess oxide above the surface.
- CMP Planarization: Remove the SiO_2 overburden using chemical reactions and mechanical abrasion.
- Selective Polishing: Use an oxide-selective slurry to remove SiO_2 while minimizing erosion of the underlying Si_3N_4 polishing stop layer.
- Endpoint Control: Stop polishing after oxide clearing, with controlled overpolishing to minimize residual oxide and nitride loss.



https://www.sciencedirect.com/topics/engineering/silicon-and-silicon-oxide

STI CMP – Removal of Overburden Oxide

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Key Process Challenges

- Oxide dishing: Excessive oxide removal within trenches.
- Nitride erosion: Loss of the Si_3N_4 stop layer.
- Non-uniformity: Uneven oxide removal across the wafer.
- Defects: Scratches, residual ceria particles, and surface contamination.

Component	Material / Chemical	Function
Film to remove	SiO_2	Overburden oxide
Stop layer	Si_3N_4	CMP stop / active-area protection
Abrasive	CeO_2 (ceria)	Oxide removal
Slurry chemistry	Water-based, pH-adjusted additives	Control oxide removal and selectivity
Additives	Dispersants, selectivity modifiers	Control particle stability and nitride removal

Tungsten (W) CMP — Chemical Reactions and Removal Mechanisms

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Tungsten CMP uses controlled oxidation, surface passivation, and mechanical abrasion to remove W overburden during contact plug and via fabrication.

1. Main Chemical Reactions

Step 1 — Tungsten Oxidation

Hydrogen peroxide (H_2O_2) oxidizes metallic tungsten to tungsten oxide (WO_3).

$$\mathbf{W + 3H_2O_2 \rightarrow WO_3 + 3H_2O}$$

Step 2 — Oxide Hydration and Dissolution

Tungsten oxide may form hydrated oxide or soluble tungstate species, depending on slurry pH. Under alkaline conditions:

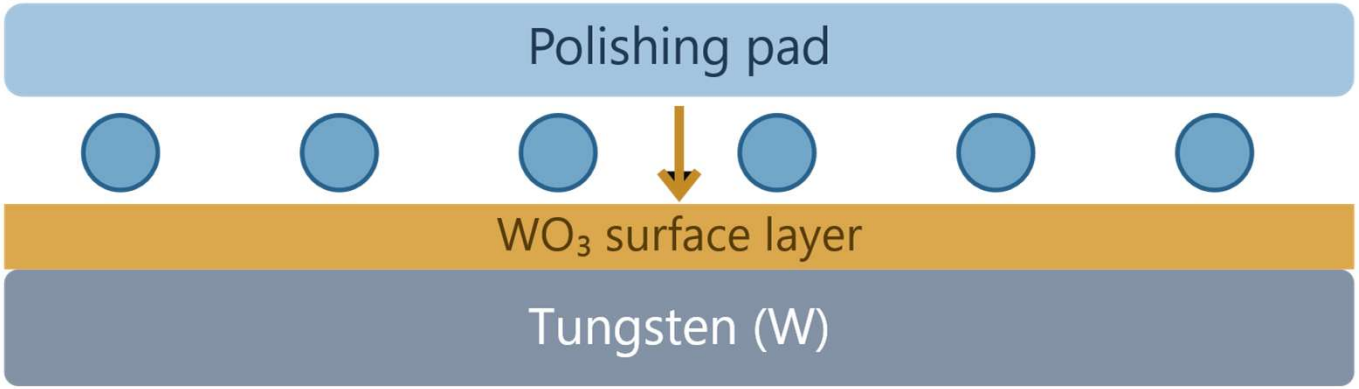
$$\mathbf{WO_3 + H_2O \rightleftharpoons H_2WO_4}$$

Step 3 — Mechanical Removal

Abrasive particles (SiO_2 or Al_2O_3) and polishing pad contact remove the oxidized surface layer, exposing fresh W for continued oxidation.

$$\mathbf{WO_3 + 2OH^- \rightarrow WO_4^{2-} + H_2O}$$

Typical Chemicals Used in Tungsten CMP



Chemical	Function
H_2O_2	Oxidizer; forms tungsten oxide
$Fe(NO_3)_3$ / Fe^{3+}	Oxidation catalyst in certain W CMP formulations
Silica (SiO_2)	Abrasive for mechanical removal
Alumina (Al_2O_3)	Alternative abrasive for W removal
HNO_3 / Organic Acids	Adjust pH and influence oxide dissolution
Corrosion Inhibitors	Suppress unwanted W dissolution and defects

Key Process Parameters & Challenges

- W dishing: Excessive tungsten removal inside contact plugs.
- Oxide erosion: Unwanted removal of surrounding dielectric.
- Corrosion and pitting: Excessive chemical dissolution or localized electrochemical reactions.
- Selectivity: Control W removal relative to Ti/TiN barrier layers and SiO_2.

Parameter	Effect on W CMP
H_2O_2 Concentration	Controls oxidation rate and surface passivation
Slurry pH	Influences WO_3 stability and tungstate dissolution
Fe^{3+} Concentration	Can accelerate oxidation through catalytic reactions
Abrasive Size / Loading	Affects removal rate and scratching
Downforce / Pad Velocity	Controls mechanical removal of the modified surface

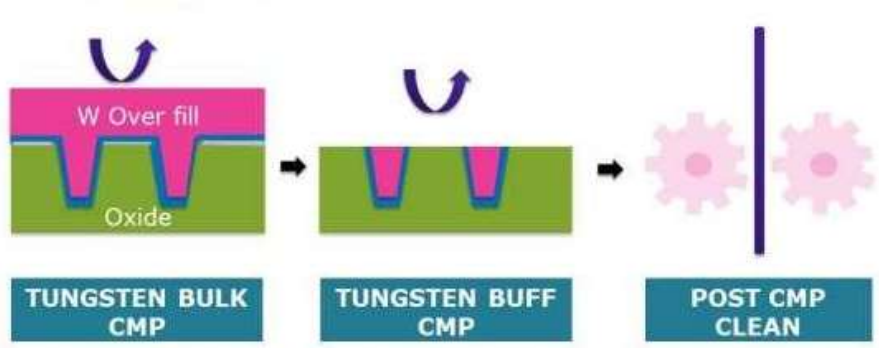
Tungsten (W) Contact Plug CMP

Removal of W Overburden

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Remove excess tungsten (W) and barrier/liner materials after contact or via filling, leaving isolated W plugs with a planar surface for subsequent semiconductor fabrication.

- **Contact/Via Formation**: Etch contact holes through dielectric layers (SiO_2) to expose underlying electrical connection regions.
- **Barrier/Liner Deposition**: Deposit Ti/TiN or TiN to promote adhesion, provide contact functionality, and support W deposition.
- **Tungsten Deposition**: Fill contact holes with W, typically using chemical vapor deposition (CVD), leaving W overburden.
- **CMP Planarization**: Remove excess W and barrier materials while retaining W inside contact holes.
- **Endpoint Control**: Stop polishing after clearing the overburden and barrier layer, minimizing W recess and dielectric erosion.



Typical W CMP Materials and Chemicals

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Key Process Challenges

- W plug recess: Excessive W removal below the surrounding dielectric surface.
- Dielectric erosion: Unwanted removal of SiO_2 around contact plugs.
- Residual W/barrier: Incomplete removal can cause electrical shorts.
- Corrosion and defects: Surface corrosion, scratches, metallic contamination, and residual abrasive particles.

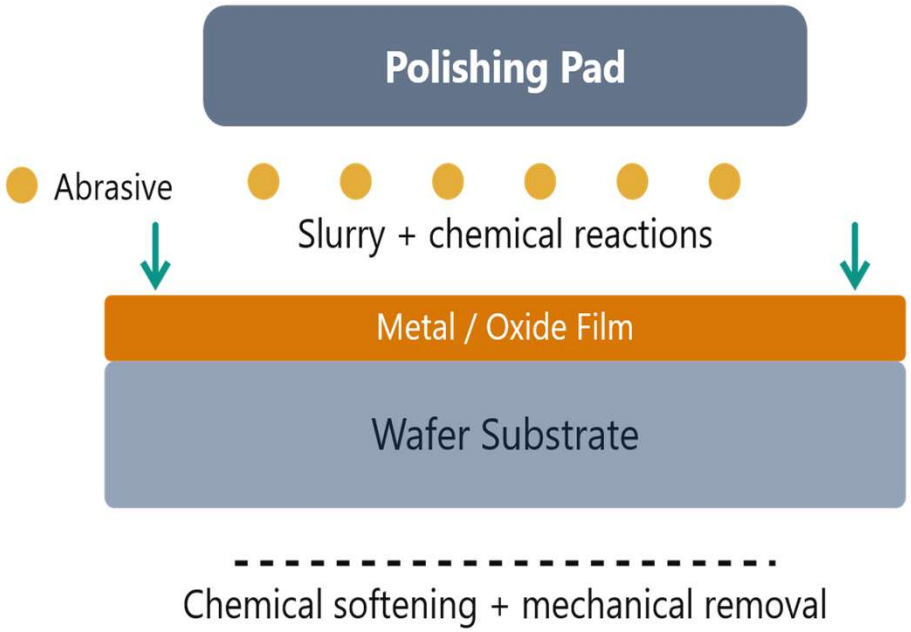
Component	Material / Chemical	Function
Film to remove	W (tungsten)	Remove metal overburden
Barrier/liner	Ti / TiN	Remove exposed liner after W clearing
Abrasive	SiO_2 (silica), Al_2O_3 (alumina)	Mechanical removal
Oxidizing agent	H_2O_2, $Fe(NO_3)_3$-based systems	Promote W surface oxidation
Catalyst	Fe-based compounds (selected slurries)	Accelerate W oxidation
pH control	Acids / pH adjusters	Control removal rate and slurry stability
Additives	Corrosion inhibitors, dispersants	Control W recess, defects, and particle stability

Chemical Mechanism of CMP

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For example, in copper CMP:

1. Oxidation: H_2O_2 oxidizes the Cu surface to form copper oxide or hydroxide species.
2. Complexation: Glycine and other complexants facilitate the dissolution of oxidized Cu.
3. Corrosion inhibition: BTA forms a protective surface film, helping control unwanted corrosion.
4. Mechanical removal: Abrasive particles and pad contact remove the reacted surface layer.
5. Planarization: The combined action preferentially removes protruding regions under appropriate process conditions.



Cu CMP – best case study for specialty chemicals

Cu CMP gives you an excellent example which can clearly see the roles of individual chemical families.

A simplified mechanism is:
Cu → oxidized/passivated Cu surface → abrasive removes modified layer → fresh Cu exposed → oxidation repeats
Hydrogen peroxide is commonly used to oxidize Cu, while complexing agents such as glycine or citrate keep Cu ions soluble. BTA is widely used to inhibit excessive Cu dissolution. ScienceDirect

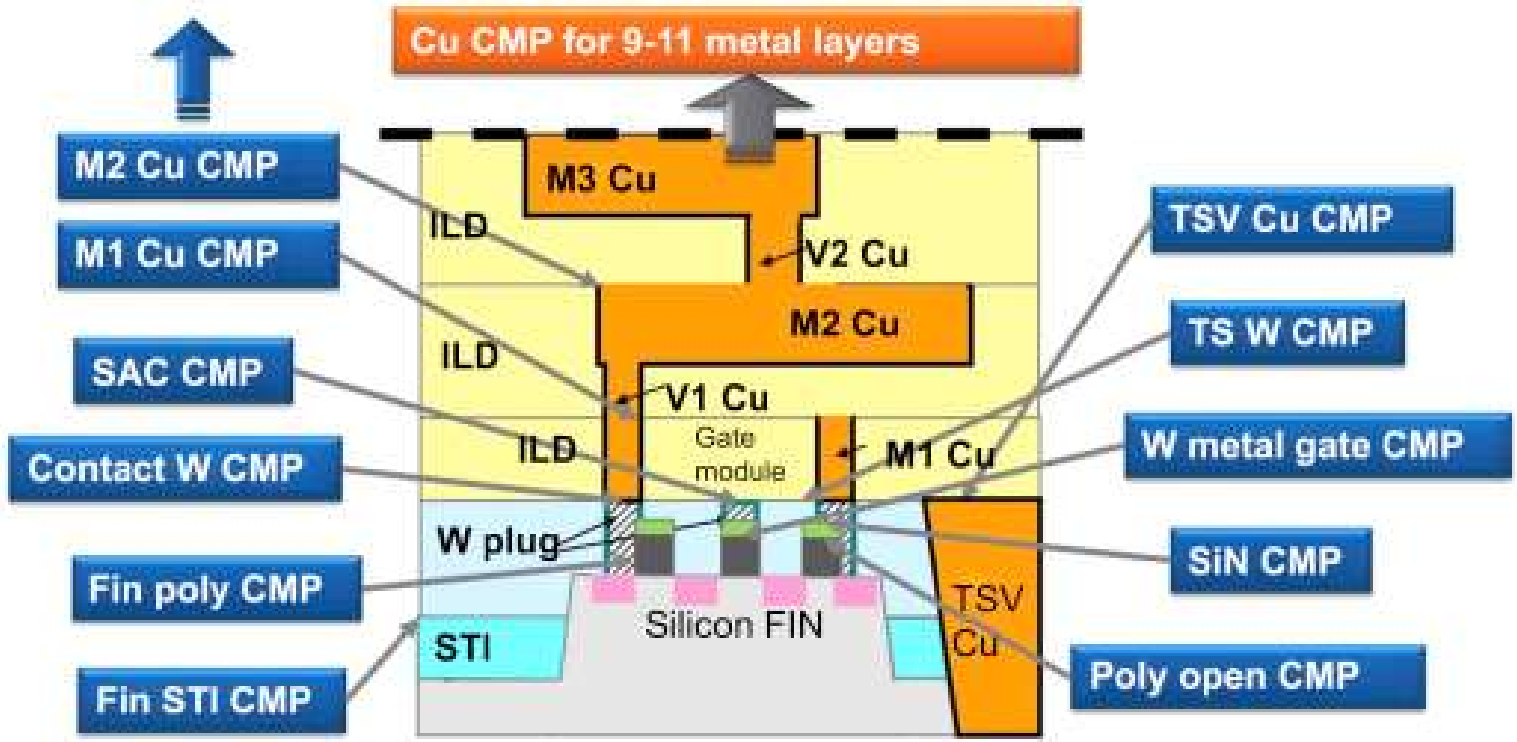
Example chemistry
Oxidation: $Cu \rightarrow Cu^{+}/Cu^{2+}$
[H_2O_2 promotes formation of Cu oxide/hydroxide species.

Complexation: $Cu^{2+} + nL \rightarrow CuL_n$
where **L = glycine, citrate, etc.**

Corrosion inhibition: *Cu + BTA → Cu–BTA protective/passivation layer*
The abrasive then mechanically disrupts/removes the surface layer at contact points.
This creates a useful conceptual competition:
Oxidation + complexation → removal versus **Passivation/inhibition → protection**

CMP performance comes from balancing the two

Cu CMP for 11 Metal Layers CMOS



https://www.sciencedirect.com/science/chapter/edited-volume/abs/pii/B9780081001653000012

Cu Interconnects (BEOL) CMP

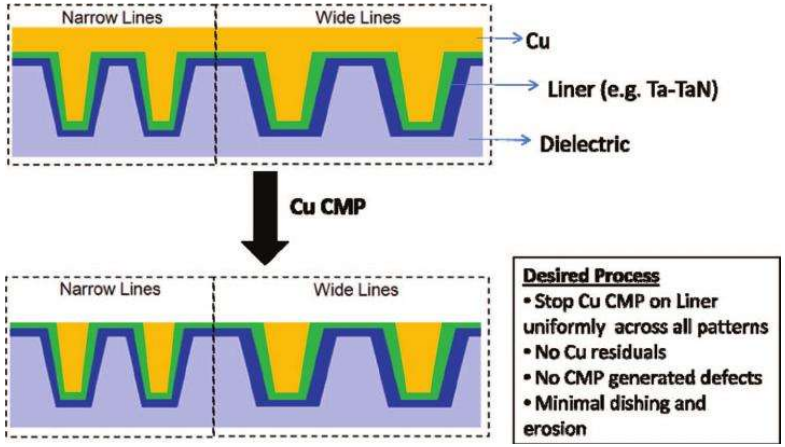
Copper Damascene Planarization

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Remove excess copper (Cu) and barrier/liner materials after damascene metallization, forming electrically isolated Cu interconnects with a planar surface for multilayer Back-End-of-Line (BEOL) integration.

Process Principle

- **Trench/Via Formation**: Pattern and etch trenches and vias in dielectric layers (SiO_2 or low-k materials).
- **Barrier/Seed Deposition**: Deposit Ta/TaN or other suitable diffusion barriers and Cu seed layers.
- **Cu Electroplating (ECP)**: Fill trenches and vias with Cu, leaving excess Cu overburden.
- **Cu CMP**: Remove excess Cu through controlled chemical oxidation, dissolution, and mechanical abrasion.
- **Barrier CMP**: Remove exposed barrier materials (e.g., Ta/TaN) while minimizing dielectric loss and Cu recess.



Typical Cu CMP Materials and Chemicals

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Key Process Challenges

- Cu dishing: Excessive Cu removal within wide trenches, causing recessed interconnect surfaces.
- Dielectric erosion: Unwanted dielectric removal, particularly in dense interconnect regions.
- Cu corrosion: Chemical attack, pitting, or galvanic corrosion during CMP and cleaning.
- Residual Cu/barrier: Incomplete removal may cause electrical shorts between interconnects.
- Surface defects: Scratches, abrasive particles, organic residues, and metallic contamination.

Component	Material / Chemical	Function
Film to remove	Cu	Remove Cu overburden
Barrier layer	Ta / TaN	Remove exposed diffusion barrier
Abrasive	SiO_2 (silica), Al_2O_3 (alumina)	Mechanical removal
Oxidizing agent	H_2O_2	Promote Cu surface oxidation
Complexing agent	Glycine, citric acid	Enhance Cu dissolution
Corrosion inhibitor	Benzotriazole (BTA)	Suppress Cu corrosion and dishing
pH adjuster	NH_4OH, KOH, organic acids	Control removal rate and slurry stability
Additives	Surfactants, dispersants	Improve slurry stability and defect control

Key CMP Process Parameters

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For semiconductor manufacturing, chemical selection should be evaluated together with:

- Material removal rate (MRR), typically nm/min
- Selectivity between target and stop layers
- Within-wafer non-uniformity (WIWNU)
- Dishing and erosion
- Surface roughness (Ra/Rq)
- Defect density, scratches, and particle contamination
- Slurry stability, pH, oxidizer decomposition, and shelf life
- Environmental, health, and safety considerations

CMP Selectivity and Endpoint Control

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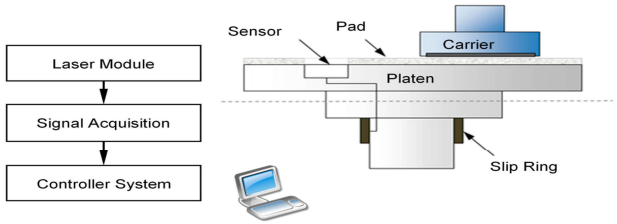
CMP selectivity determines the relative removal rates of different materials, while endpoint control determines when polishing should stop to achieve the required film thickness and minimize over-polishing.

CMP Selectivity
RR = Removal Rate (nm/min or μm/min)

$$S_{A/B} = \frac{RR_A}{RR_B}$$

CMP Application	Selectivity Requirement	Purpose
STI CMP	High SiO_2 : Si_3N_4	Stop on Si_3N_4 and minimize nitride loss
W Contact CMP	Controlled W : Ti/TiN : SiO_2	Remove W overburden while protecting plugs and dielectric
Cu CMP	Controlled Cu : Barrier : Dielectric	Minimize Cu dishing and dielectric erosion
ILD CMP	Controlled oxide removal	Achieve target thickness and uniform planarization

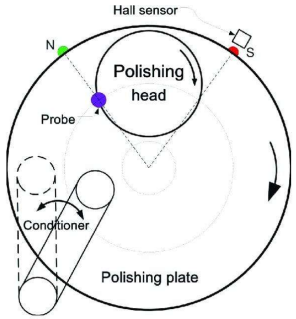
CMP Endpoint Detection Methods



Optical Endpoint Detection
Monitors reflected light intensity or spectral changes to detect film thickness and material transitions. Common in dielectric CMP and compatible metal CMP applications.



Motor Current / Friction Monitoring
Detects changes in polishing friction or motor torque when the exposed surface material changes.

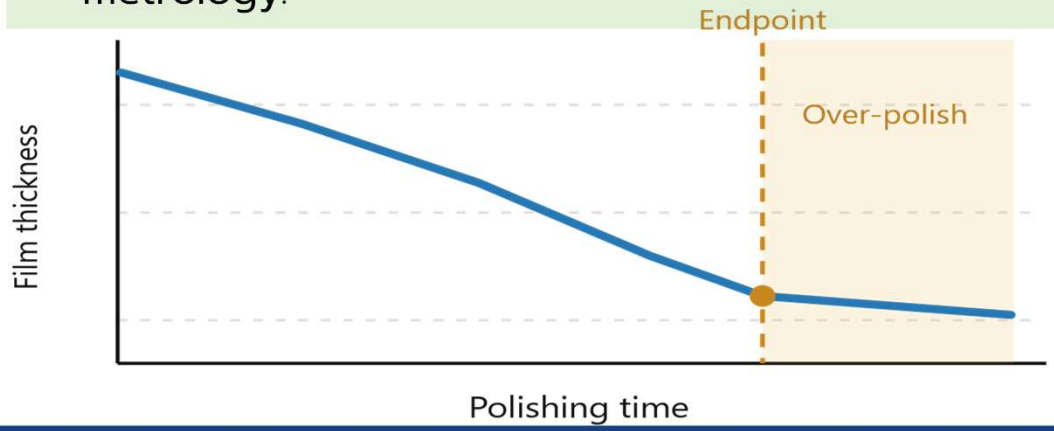


Eddy Current Monitoring
Measures changes in conductive metal film thickness, particularly useful for Cu CMP.

CMP Endpoint Detection Methods

Time-Based Endpoint
Uses calibrated removal rate and polishing time. Simple but sensitive to process drift and wafer-to-wafer variations.

- **Under-polishing**: Residual metal or dielectric overburden remains.
- **Over-polishing**: Excessive dishing, erosion, and stop-layer loss.
- **Endpoint optimization**: Combine in-situ detection, calibrated over-polish, and post-CMP metrology.



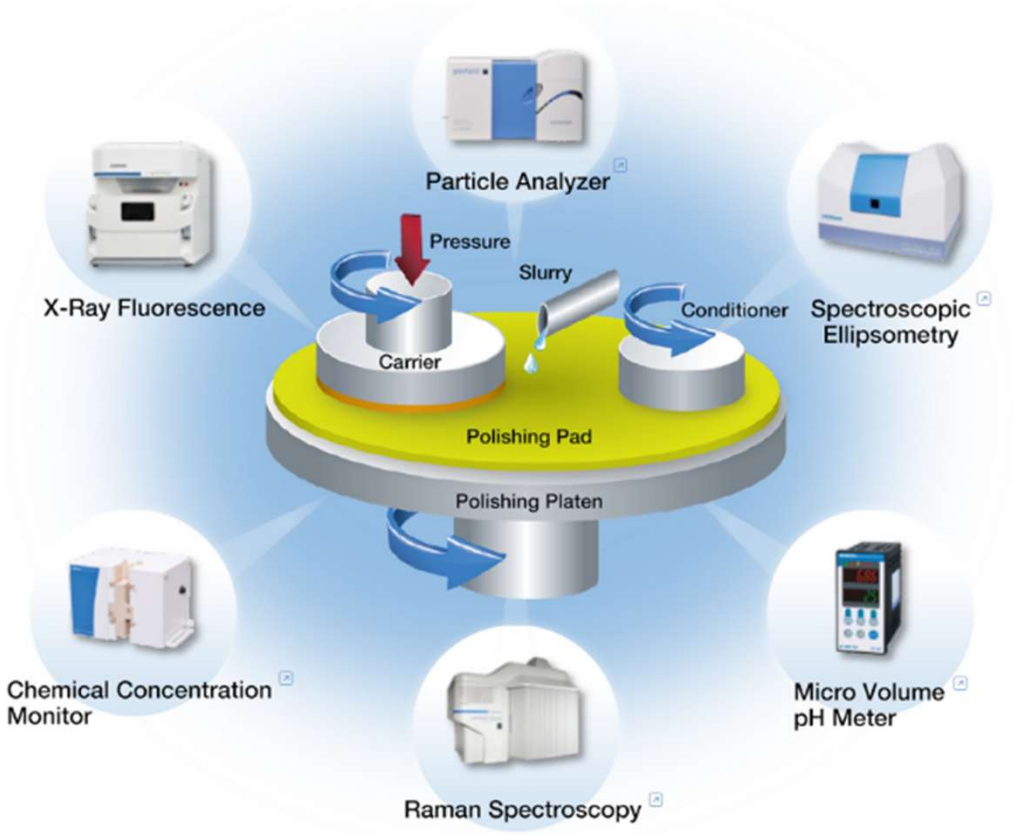
Key Process Challenges
- Selectivity vs. removal rate: High removal rate must be balanced against stop-layer protection.
- Endpoint accuracy: Influenced by pattern density, film thickness variation, and sensor sensitivity.
- Wafer uniformity: Different wafer regions may reach endpoint at different times.
- Process control: Slurry chemistry, pad conditioning, and polishing parameters must remain stable.

Important CMP defects to connect to chemistry

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Defect/problem	Chemical/process relationship
Scratches	Large/agglomerated abrasive particles
Dishing	Excessive Cu removal / overpolish
Erosion	Pattern-density/selectivity effects
Corrosion	Insufficient inhibition
Galvanic corrosion	Cu/barrier electrochemical potential difference
Residue	Slurry/additive/metal residues
Particle contamination	Slurry stability / post-CMP cleaning
Low removal rate	Excessive passivation or weak oxidation
High removal rate	Aggressive oxidation/complexation
Non-uniformity	Slurry transport + pad + chemical kinetics

Analysis tools for CMP process



- Particle Size and Count Measurement
- Zeta Potential Measurement
- CMP Slurry Composition Monitoring
- ph Monitoring
- Chemical Concentration Measurement
- Thin Film Thickness Measurement
- Trench Profile Measurement

https://www.horiba.com/ind/semiconductor/applications/chemical-mechanical-planarization-cmp/

Why wafer cleaning matters

Cleaning → the most critical yield enablers in semiconductor manufacturing

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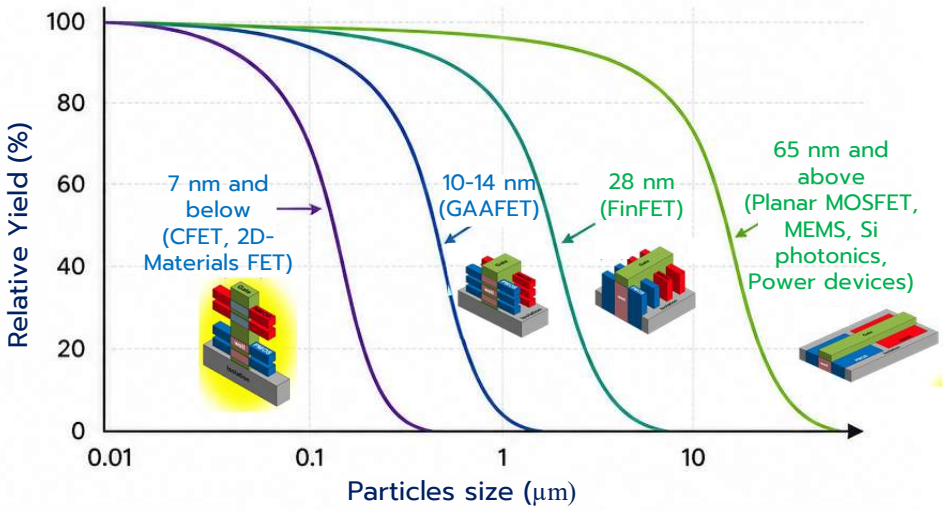
~30-40%
Of process steps involve cleaning

Cleaning impacts:

- Yield
- Reliability
- Defectivity
- Device performance

Defect sensitivity increases dramatically at advanced nodes

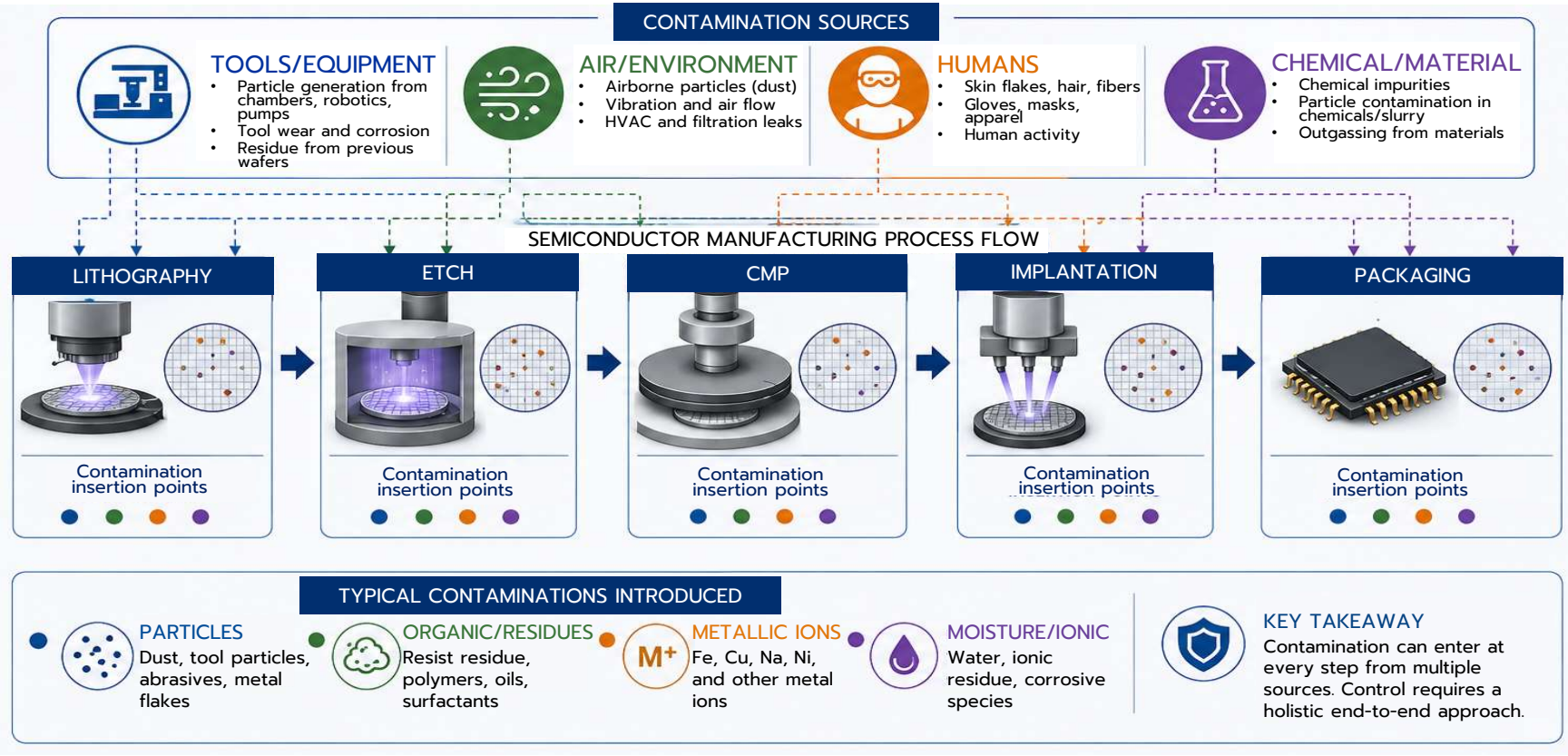
YIELD vs PARTICLE SIZE RELATIONSHIP



At advanced nodes, even sub-10 nm particles can cause yield loss

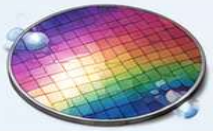
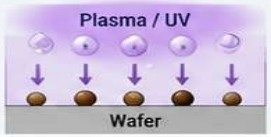
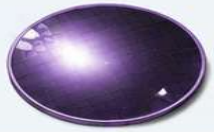
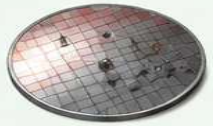
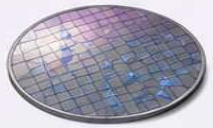
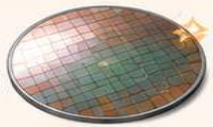
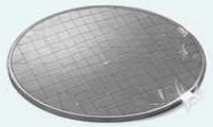
Sources of Contamination

Contamination can be introduced throughout the manufacturing flow



Wafer Cleaning Classification

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Category	Examples	Principle / Mechanism	Typical Targets / Applications
Wet cleaning	**RCA, DHF** 	Chemical solution dissolves, complex, or oxidize contaminants; surfactants and agitation help lift particles from the surface 	• Organic residues • Metal ions • Native oxide • Particles 
Dry cleaning	**Plasma, UV/Ozone** 	Reactive species (ions, radicals, photons) break down or volatilize organic films and residues without liquid contact 	• Organic film • Photoresist residue • Polymer • VOCs 
Mechanical cleaning	**Scrubber** 	Physical brushing and chemical action remove particles through mechanical force and solution flow 	• Partibles • Heavy contamination • Metallic particles 
Acoustic cleaning	**Megasonic** 	High-frequency sound waves (0.5-2 MHz) create cavitation bubbles in the liquid that collapse and dislodge particles 	• Fine particles • Sub-micron particles • High aspect ratio structures 
Vapor cleaning	**HF vapor** 	Vapor-phase chemicals react with surface contaminants and residues; by-products are pumped away 	• Native oxide • Stiction layers • Sacrificial oxide 
CO_2 Supercritical cleaning	**CO_2 (Supercritical)** 	Supercritical CO_2 has gas-like diffusivity and liquid- like density- dissolves organics and removes contaminants without liquid-vapor interface 	• Stiction-free drying • Organic residues • Micro-mechanical structures (MEMS) 

Post-CMP Cleaning

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Purpose: Remove slurry particles, metal ions, organic residues, and polishing byproducts without damaging wafer surfaces.

Cleaning mechanisms

- Chemical dissolution and chelation
- Electrostatic particle repulsion
- PVA brush mechanical removal
- Megasonic-assisted particle detachment
- UPW rinsing and drying

Key chemicals: NH_4OH, organic acids, chelators, surfactants, formulated corrosion inhibitors, and UPW.

Major challenges: Particle redeposition, Cu/W corrosion, brush-induced defects, watermarks, and cross-contamination.

Key message: Effective post-CMP cleaning requires optimizing chemical formulation and mechanical cleaning together to achieve low defectivity and high device yield.

Post-CMP Cleaning

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Remove residual slurry particles, organic contaminants, metallic impurities, and chemical residues from wafer surfaces after CMP.

The primary objective is to achieve a particle-free, residue-free, corrosion-free, and defect-free wafer surface before subsequent semiconductor fabrication processes.

Contamination type	Typical sources	Potential problems
Abrasive particles	SiO_2, CeO_2, Al_2O_3 slurry	Defects, scratches, electrical failures
Metallic contamination	Cu, W, Fe, Co and other metals	Leakage current, reliability degradation
Organic residues	Surfactants, corrosion inhibitors, dispersants	Poor adhesion, surface defects
Chemical residues	Oxidizers, complexing agents, pH adjusters	Corrosion, surface instability
Polishing byproducts	Metal oxides, hydroxides, reaction products	Surface contamination, defects
Pad debris	Polyurethane polishing pad	Particle defects, surface contamination

Post-CMP Cleaning

Key challenge: CMP combines chemical and mechanical material removal, so post-CMP cleaning must address both chemical residues and physically attached particles without damaging the underlying films.

PVA Brush Scrubbing
Uses soft, porous polyvinyl alcohol brushes and cleaning chemicals to detach slurry particles and residues. Widely used in single-wafer post-CMP cleaning.

Megasonic Cleaning
Uses high-frequency acoustic energy to detach fine particles without direct brush contact. Useful for sensitive surfaces, although acoustic damage must be controlled.

Spray / Spin Cleaning
Uses chemical or UPW spray on a rotating wafer to rinse contaminants and residues. Often combined with chemical treatment and spin drying.

Megasonic Cleaning

High-frequency acoustic energy for superior particle removal

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HOW MEGASONIC CLEANING WORKS (MECHANISM)

1. Acoustic Wave Generation

High-frequency sound waves propagate through the liquid

2. Acoustic Streaming (Bubble Streaming)

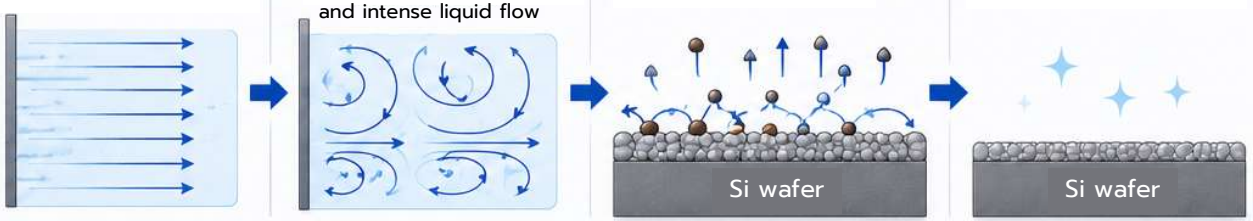
The acoustic waves create microstreaming and intense liquid flow

3. Particle Detachment & Removal

Streaming, microjets, and shear forces dislodge particles from the surface

4. Clean Surface

Particles are removed and fresh solution reaches the surface



MEGASONIC vs ULTRASONIC

	Megasonic (High Frequency)	Ultrasonic (Low Frequency)
Frequency	0.5-5 MHz	20 -100 kHz
Acoustic mechanism	Acoustic streaming (Non-cavitation)	Cavitation (Bubble collapse)
Bubble activity	Very fine, stable bubbles (No violent collapse)	Large bubbles, Violent collapse
Cleaning action	Microstreaming, shear, microjets	Shock waves, bubble collapse
Surface damage risk	Very Low	higher
Best for	Advanced nodes, fragile structure, low-k, EUV	General cleaning, particles, bulk contaminants
Particle removal	Excellent for fine particles (< 0.1 μm)	Good for large particles (> 0.1 μm)

MECHANISM IN DETAIL

Acoustic Streaming

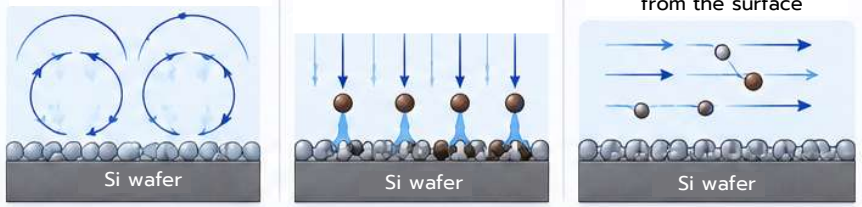
High-frequency waves generate circulating flow in the liquid

Microjets

Localized high-velocity jets remove particles from surface features

Shear Forces

Shear stress from streaming lifts and carries particles away from the surface

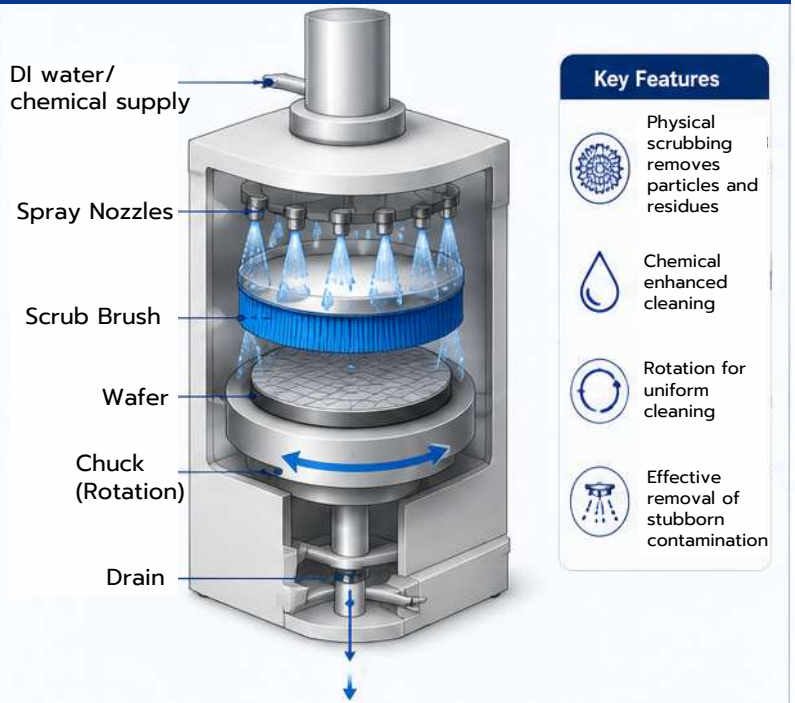


Wafer scrubber & Brush for cleaning process

High-efficiency physical cleaning for particle and residue removal

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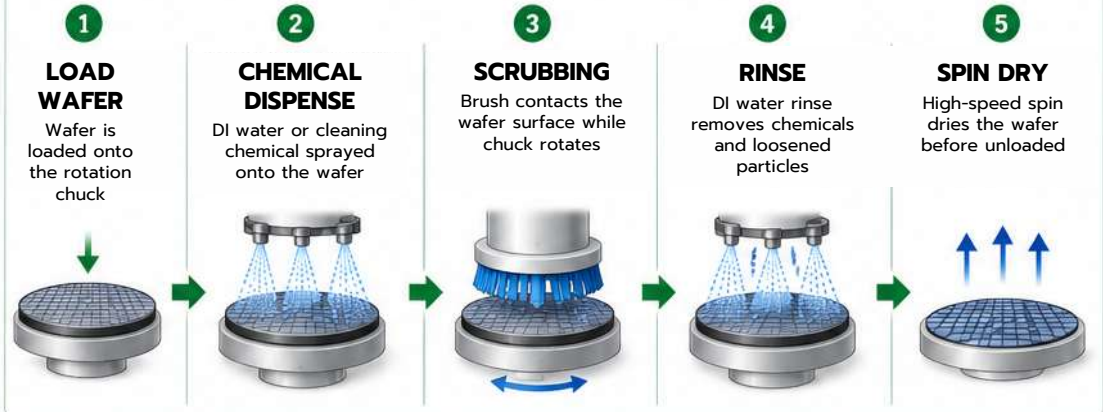
1. WAFER SCRUBBER



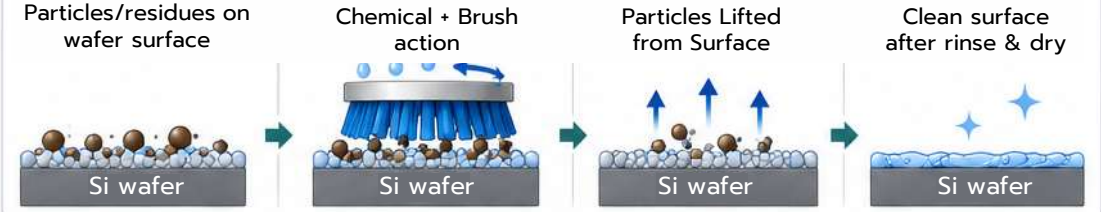
Key Features

- Physical scrubbing removes particles and residues
- Chemical enhanced cleaning
- Rotation for uniform cleaning
- Effective removal of stubborn contamination

2. CLEANING PROCESS



3. HOW IT WORKS

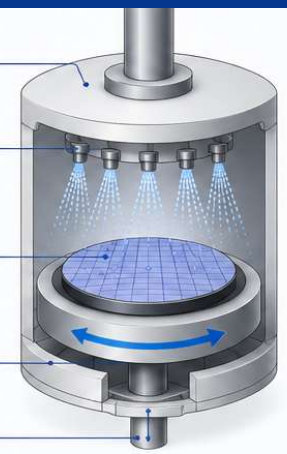


Single-wafer spin cleaning

High-performance cleaning with uniform chemical distribution

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1. SPIN CLEANING CHAMBER



Spray Nozzles
Uniform chemical coverage

Chemical Jets
High-velocity jets deliver chemicals uniformly

Rotating Wafer
Wafer spins at 500-3000 rpm for effective cleaning

Vacuum Chuck
Holds wafer securely during rotation

Drain
Removes used chemicals and by-products

2. CLEANING PROCESS FLOW



1. Chemical Dispense	2. Spin Cleaning	3. Rinse	4. Spin Dry	5. Complete
Chemical are sprayed onto the rotating wafer	Rotation and chemical action remove particles and organic residues	DI water rinse removes chemicals and by-products	High-speed spin removes water from the water surface	Clean, dry wafer ready for the next process

3. BEFORE/ AFTER CLEANING



BEFORE CLEANING
Particles, organics, and residues presents

AFTER CLEANING
Clean, particle-free, residue-free surface

4. KEY BENEFITS

- Excellent particle and contamination removal
- High cleaning uniformity and repeatability
- Improves yield and device reliability
- Low chemical consumption and waste
- Ideal for advanced node

5. TYPICAL APPLICATIONS



- Post-CMP cleaning
- Photoresist removal
- Front-end cleaning
- General wafer cleaning

Recommended Cleaning Chemistry by CMP Application

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CMP application	Typical cleaning chemistry	Main cleaning objective	Critical concern
STI oxide CMP	Alkaline cleaners, specialized ceria-removal chemistry	Remove CeO_2 and oxide residues	Strong ceria adhesion
Dielectric CMP	Dilute NH_4OH, surfactants, formulated alkaline cleaners	Remove silica particles	Particle redeposition
Tungsten plug CMP	Organic acids, chelators, formulated pH-controlled cleaners	Remove W/Fe residues and silica	W corrosion, oxide attack
Cu interconnect CMP	Organic acids, chelators, surfactants, controlled inhibitors	Remove Cu residues, BTA and particles	Cu corrosion, galvanic attack
Barrier CMP (Ta/TaN)	Formulated acidic or alkaline cleaners	Remove barrier residues and particles	Mixed-material compatibility
SiC CMP	Formulated cleaners, surfactants and chelators	Remove abrasives and oxidized residues	Surface defects and scratches

Critical Process Parameters for Post-CMP Cleaning

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Parameter	Why it matters
Cleaning chemistry / pH	Controls particle surface charge, solubility and corrosion
Chemical concentration	Balances cleaning efficiency and material compatibility
Brush pressure / rotation speed	Controls particle removal and mechanical damage
Megasonic power / frequency	Influences particle removal and pattern damage
Cleaning time	Controls residue removal and potential surface attack
UPW quality	Prevents ionic and particulate recontamination
Drying conditions	Prevents watermarks and residual stains
Brush condition	Prevents cross-contamination and particle redeposition

Chemical Safety and Waste Management in CMP

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Chemical / Material	Major Hazards	Safety Precautions
H_2O_2 (Hydrogen Peroxide)	Strong oxidizer; decomposition and pressure buildup	Segregate from organics, metals, and incompatible chemicals
KOH / NH_4OH	Corrosive alkaline chemicals; NH_3 vapor from ammonium hydroxide	Chemical-resistant PPE and local exhaust
HNO_3 / Acidic Slurries	Corrosion, burns, potentially hazardous fumes	Acid-compatible storage and handling
Silica / Ceria / Alumina Slurry	Particle exposure, irritation, sedimentation	Closed chemical delivery and appropriate filtration
Metal-Containing Waste	Cu, W, Co, Ni contamination	Segregated collection and metal removal treatment
Organic Additives	Chemical exposure, increased wastewater COD	Appropriate storage and wastewater treatment

Key Waste Management Considerations

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- Waste segregation: Prevent incompatible chemicals from mixing, especially oxidizers with organic-rich or reactive waste streams.
- Wastewater monitoring: Track pH, TSS, COD, dissolved metals, and other site-specific discharge parameters.
- Water reuse: Apply suitable filtration and polishing treatment where technically and economically feasible.
- Regulatory compliance: Follow applicable Thai industrial wastewater, hazardous waste, and occupational safety requirements.

Waste Type	Typical Treatment
Abrasive particles	Coagulation, flocculation, sedimentation, membrane filtration
Dissolved metal ions	Chemical precipitation, adsorption, ion exchange
Acidic / Alkaline wastewater	Controlled pH neutralization
Organic compounds	Biological treatment or advanced oxidation, as appropriate
CMP sludge	Dewatering, characterization, and licensed waste disposal

CMP Chemicals and Post-CMP Cleaning – Key Takeaways

- CMP is essential for semiconductor integration, enabling planarization of STI, dielectric films, W contacts, and Cu interconnects.
- CMP slurry is a chemically engineered formulation combining abrasives, oxidizers, complexants, inhibitors, and other additives.
- Different materials require different slurry chemistries to achieve controlled removal rates, selectivity, and defect performance.
- Post-CMP cleaning is essential for yield and reliability, removing particles, organic residues, and metallic contamination while protecting sensitive films.
- CMP performance requires integrated control of chemistry, polishing mechanics, cleaning, metrology, and contamination.

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